

## Participación en Congresos

| Nombre del Congreso                                                                    | Título del Trabajo                                                                                                                    | Tipo de Participación         | País           | Fecha     |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|-----------|
| Encuentro Nacional de Computación, EUREKA ENC 2024                                     | Human Activity Recognition on the Edge with WiFi Channel State Information                                                            | Ponencia                      | México         | 24-Aug-24 |
| XVIII jornadas iberoamericanas de ingeniería de software e ingeniería del conocimiento | Revisión de literatura de elementos de minería de procesos para la minería de procesos de software                                    | Ponencia                      | México         | 22-Nov-23 |
| Encuentro nacional de computación, ENC 2023                                            | CSI crowd-counting: an experimental study using machine learning and deep learning algorithms                                         | Ponencia                      | México         | 11-Sep-23 |
| Encuentro nacional de computación, ENC 2023                                            | Convolutional neural networks applied to classification of nanoparticles and nanotubes images                                         | Ponencia                      | México         | 11-Sep-23 |
| 10th international conference in software research and innovation, CONISOFT 2022       | Process mining model integrated with control flow, case, organizational and time perspectives in a software development project       | Ponencia                      | México         | 24-Oct-22 |
| Encuentro nacional de computación, ENC 2022                                            | Detection of atypical data in point cloud of technical vision system using digital filtering                                          | Ponencia                      | México         | 24-Aug-22 |
| Encuentro nacional de computación, ENC 2022                                            | Intervention of knowledge flows in software process mining projects                                                                   | Ponencia                      | México         | 24-Aug-22 |
| Encuentro nacional de computación, ENC 2022                                            | Nanoparticles diameter characterization using image analysis methodology                                                              | Ponencia                      | México         | 24-Aug-22 |
| 2022 IEEE 31st international conference on artificial intelligence, MICA 2022          | Obstacle Coordinates transformation from TVS body-frame to AGV Navigation-Frame                                                       | Presentación Artículo Extenso | Estados Unidos | 1-Jun-22  |
| Encuentro nacional de computación, ENC 2021                                            | BREML: A breathing rate estimator using wi-fi channel state information and machine learning                                          | Ponencia                      | México         | 9-Aug-21  |
| Encuentro nacional de computación, ENC 2021                                            | WIFI-ALR wifi alerts software for human movements using machine learning algorithms                                                   | Ponencia                      | México         | 9-Aug-21  |
| Encuentro nacional de computación, ENC 2020                                            | Software para emisión de alertas basadas en detección de movimiento por señales wifi                                                  | Ponencia                      | México         | 24-Aug-20 |
| Encuentro nacional de computación, ENC 2020                                            | Educational data mining for finding patterns in first semester engineering students outcomes to identify potential success or failure | Ponencia                      | México         | 24-Aug-20 |
| Encuentro nacional de computación, ENC 2020                                            | Human activity recognition from wrist wearable device sensors using data mining techniques                                            | Ponencia                      | México         | 24-Aug-20 |
| 2020 IEEE 29th international symposium on industrial electronics (ISIE)                | A lean convolutional neural network for vehicle classification                                                                        | Presentación Artículo Extenso | Países Bajos   | 17-Jun-20 |
| 2020 IEEE 29th international symposium on industrial electronics (ISIE)                | Classification of vehicle images through deep neural networks for camera view position selection                                      | Presentación Artículo Extenso | Países Bajos   | 17-Jun-20 |
| IECON 2019 45th annual conference of the IEEE industrial electronics society           | An MHD stroller 2D velocity profile measurement validation through a machine vision system                                            | Ponencia                      | Portugal       | 14-Oct-19 |

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|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|--------------|
| VI Latin American conference on biomedical engineering                                | Hand movement detection from surface electromyography signals by machine learning techniques                                                           | Ponencia                      | México         | 3-Oct-19     |
| Congreso internacional de investigación académica, Journals Chetumal 2009             | Caracterización de la satisfacción del cliente y calidad del software en pymes en Baja California                                                      | Presentación Artículo Extenso | México         | 22-May-19    |
| 33rd AAAI conference on artificial intelligence, AAAI 19                              | Domain adaptation for human fall detection using wifi channel state information                                                                        | Presentación Artículo Extenso | España         | 27-Jan-19    |
| IEEE international conference on pervasive computing and communications workshops     | A lifeling platform towards detecting negative emotions in everyday life using wearable devices                                                        | Ponencia                      | Grecia         | 19-Mar-18    |
| Congreso internacional de investigación                                               | La satisfacción del cliente en la utilización del software de las pequeñas y medianas empresas (pymes) en Baja California                              | Presentación Artículo Extenso | México         | 8-Nov-17     |
| 2017 IEEE international symposium on industrial electronics, ISIE 2017                | Machine vision system to measuring the velocity field in a fluid by particle image velocimetry                                                         | Presentación Artículo Extenso | Reino Unido    | 18-Jun-17    |
| 2017 IEEE international symposium on industrial electronics, ISIE 2017                | Virtual angle measurement through FPGA data processing                                                                                                 | Presentación Artículo Extenso | Reino Unido    | 18-Jun-17    |
| 7 conferencia iberoamericana de complejidad, informática y cibernética                | Análisis del desarrollo del software en pequeñas y medianas empresas (pymes) del estado de Baja California, México                                     | Presentación Artículo Extenso | Estados Unidos | 19-Mar-17    |
| 11 congreso colombiano de computación                                                 | Process mining in software process improvement, a systematic literature review                                                                         | Ponencia                      | Colombia       | 28-Sep-16    |
| 18th international conference on artificial intelligence, WORLDCOMP 2016              | OCR for unreadable damaged characters on PCBS using GSC algorithm and KNN classifier                                                                   | Presentación Artículo Extenso | Estados Unidos | 18-Jun-16    |
| II simposio de nanociencias y nanomateriales                                          | Dinámica molecular vs simulación directa montecarlo. Generación de trayectorias y análisis de propiedades                                              | Poster                        | México         | 10-Jun-16    |
| 2do simposio de nanomateriales y nanomateriales                                       | Métodos de simulación montecarlo y su empleo en ingeniería de materiales                                                                               | Poster                        | México         | 6-Jun-16     |
| 2015 international conference on computational science and computational intelligence | OCR for unreadable damaged characters on PCBS using principal component analysis and bayesian discriminant functions                                   | Presentación Artículo Extenso | Estados Unidos | 7-Dec-15     |
| 41st annual conference of the IEEE industrial electronics society, IECON 2015         | Outlier mining of a vision sensing database for SVM regression improvement                                                                             | Presentación Artículo Extenso | Japón          | 9-Nov-15     |
| Congreso internacional de investigación e innovación en ingeniería de software        | Modelación matemática de la respuesta de un biosensor de glucosa-oxidada de 2da generación mediante técnicas de reconocimiento estadístico de patrones | Presentación Artículo Extenso | México         | 1-Jan-15     |
| 13th Mexican international conference on artificial intelligence                      | Bioinformatics and medical applications                                                                                                                | Moderador                     | México         | 17-Nov-14    |
| Encuentro Nacional de Computación, ENC 2014                                           | La exteriorización de activos de conocimiento en mejora de procesos software                                                                           | Presentación Artículo Extenso | México         | 3-Nov-14     |

| <b>Nombre del Congreso</b>                                                                               | <b>Título del Trabajo</b>                                                                                           | <b>Tipo de Participación</b>  | <b>País</b>    | <b>Fecha</b> |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------|--------------|
| IX congreso de la asociación española de climatología                                                    | Modelación de temperaturas del aire a 850 MB en el caso de la onda cálida de julio de 2006 en el noroeste de México | Presentación Artículo Extenso | España         | 28-Oct-14    |
| 2014 IEEE 23rd international symposium on industrial electronics, ISIE 2014                              | Machine vision supported by artificial intelligence applied to rotary mirror scanners                               | Presentación Artículo Extenso | Turquía        | 1-Jun-14     |
| 13th Mexican international conference on artificial intelligence                                         | Non-deterministic local search methods for feature selection: an experimental study                                 | Presentación Artículo Extenso | México         | 1-Jan-14     |
| World congress on engineering and computer science, WCECS 2013                                           | Digital signal processing on optoelectronic for SHM                                                                 | Presentación Artículo Extenso | Estados Unidos | 23-Oct-13    |
| 10th Mexican International conference on artificial intelligence, MICAI 2011                             | Microarray gene subset selection in amyotrophic lateral sclerosis classification                                    | Presentación Artículo Extenso | México         | 26-Nov-11    |
| Eigth international meeting on computational intelligence methods for bioinformatics and biostatistics   | Feature selection in proton magnetic resonance spectroscopy data of brain tumors                                    | Presentación Artículo Extenso | Italia         | 30-Jun-11    |
| The 2010 world congress in computer science, computer engineering, and applied computing, worldcomp 2010 | Differentiation of glioblastomas and metastases using 1h-mrs spectral data                                          | Presentación Artículo Extenso | Estados Unidos | 1-Jan-10     |
| 8th Mexican international conference on artificial intelligence, MICAI 2009                              | Using machine learning techniques to explore 1h-mrs data of brain tumors                                            | Poster                        | México         | 1-Jan-09     |
| XXII cingreso nacional y VIII congreso internacional de informática y computación, ANEI 2009             | Machine learning methods for classifying normal vs. Tumorous tissue with special data                               | Presentación Artículo Extenso | México         | 1-Jan-09     |
| 16th european symposium on artificial neural networks, ESANN 2008                                        | Feature selection in proton magnetic resonance spectroscopy for brain tumor classification                          | Presentación Artículo Extenso | Bélgica        | 1-Jan-08     |
| II Congreso español de informática. IV taller de minería de datos y aprendizaje, TAMIDA 2007             | Un algoritmo para el cálculo de la relevancia entrópica multivariada y su uso en la selección de variables          | Presentación Artículo Extenso | España         | 1-Jan-07     |
| Simposium de inteligencia computacional, SICO 2007                                                       | A thermodynamic algorithm for feature selection                                                                     | Presentación Artículo Extenso | España         | 1-Jan-07     |
| 2do. Congreso internacional de educación                                                                 | Implementación del nuevo examen psicométrico de la U.A.B.C. En computadora                                          | Ponencia                      | México         | 1-Jan-01     |
| V congreso nacional de ciencias ambientales                                                              | Técnicas de computación avanzadas en la educación ambiental.                                                        | Ponencia                      | México         | 1-Jan-00     |

EL INSTITUTO TECNOLÓGICO DE CD. MADERO Y  
LA SOCIEDAD MEXICANA DE CIENCIA DE LA COMPUTACIÓN  
OTORGAN EL PRESENTE

# RECONOCIMIENTO

A

**Conrado García Reyes, Jesús Albany Armenta García,  
Félix F. González Navarro, Jesús Caro, Jorge Eduardo Ibarra Esquer**

Por haber presentado el artículo:

**Human Activity Recognition on the Edge with Wi-Fi Channel State Information**

en el Encuentro Nacional de Computación, Eureka ENC 2024, llevado a cabo del 23 al 25 de  
septiembre en Cd. Madero, Tamaulipas.



**Dra. Yasmín Hernández Pérez**  
Presidenta de la SMCC



**Dra. Laura Cruz Reyes**  
Organizadora Eureka ENC 2024



**Dra. Claudia Gómez Santillán**  
Organizadora Eureka ENC 2024





Otorgan la presente

# CONSTANCIA

a

**Silvia Jaqueline Urrea-Contreras, Brenda L. Flores-Rios, Angélica Astorga-Vargas,  
Jorge E. Ibarra-Esquer, Félix F. González-Navarro, Iván A. García Pacheco**

por su valiosa participación con la presentación de la **Ponencia** titulada

**Revisión de Literatura de Elementos de Minería de Procesos para la Minería  
de Procesos de software**

en el contexto de las XVIII Jornadas Iberoamericanas de Ingeniería de Software e  
Ingeniería del Conocimiento organizadas por la Facultad de Matemáticas de la  
Universidad Autónoma de Yucatán, del 22 al 23 de noviembre de 2023 en la ciudad de  
Mérida, Yucatán, México.

Dr. Raúl Antonio Aguilar Vera  
Coordinador General JIISIC 2023

M. en C. Ernesto Antonio Guerrero Lara  
Director de la Facultad de Matemáticas



El Centro de Investigación en Matemáticas A.C.,  
el Instituto Nacional de Astrofísica, Óptica y Electrónica,  
la Red Temática en Inteligencia Computacional Aplicada  
y la Sociedad Mexicana de Ciencia de la Computación  
otorgan la presente

# CONSTANCIA

a:

Miguel A. Torres Cerda, Félix F. González Navarro,  
Jesús Caro y Jesús Albany Armenta García

Por su artículo “*CSI crowd-counting: An experimental study using Machine Learning and Deep Learning Algorithms*” presentado en la edición conjunta del ENCuentro Nacional de Computación ENC2023  
y el XI SEMINARIO Y ESCUELA NACIONAL DE APRENDIZAJE E INTELIGENCIA COMPUTACIONAL  
llevada a cabo del 11 al 15 de septiembre de 2023



Dra. Yasmín Hernández Pérez  
Presidenta Comité Organizador  
ENC 2023  
CENIDET - SMCC - RedICA



Dra. Alicia Morales Reyes  
Co-Presidenta Comité Organizador  
XI SENAIC  
INAOE - RedICA

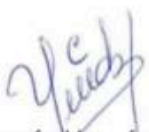
El Centro de Investigación en Matemáticas A.C.,  
el Instituto Nacional de Astrofísica, Óptica y Electrónica,  
la Red Temática en Inteligencia Computacional Aplicada  
y la Sociedad Mexicana de Ciencia de la Computación  
otorgan la presente

# CONSTANCIA

a:

Luis A. Quintero, Jesús Caro, Félix F. González  
Navarro, Mario Curiel, Oscar Pérez y Nicola Radnev  
Nedev

Por su artículo “*Convolutional neural networks applied to classification of nanoparticles and nanotubes images*” presentado en la edición conjunta del ENCuentro Nacional de Computación ENC2023 y el XI SEMINARIO Y ESCUELA NACIONAL DE APRENDIZAJE E INTELIGENCIA COMPUTACIONAL llevada a cabo del 11 al 15 de septiembre de 2023



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ENC 2023  
CENIDET - SMCC - RedICA



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Co-Presidenta Comité Organizador  
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**Universidad Tecnológica Bilingüe Internacional y Sustentable de Puebla**

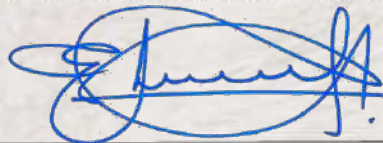
Acknowledges:

**Silvia J Urrea Contreas, Brenda L Flores-Rios, María Angélica Astorga-Vargas, Jorge Eduardo Ibarra Esquer, Felix F. Gonzalez-Navarro, Ivan Garcia, Carla Leninca Pacheco Agüero**

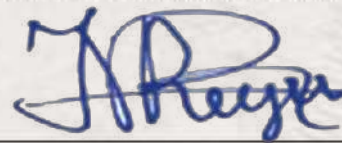
For present the article

**“Process Mining Model Integrated with Control Flow, Case, Organizational and Time Perspectives in a Software Development Project”**

For the 10th International Conference in Software Engineering Research and Innovation (CONISOFT 2022) held at Universidad Tecnológica Bilingüe Internacional y Sustentable de Puebla, October 24-28, 2022 in Ciudad Modelo, San José Chiapa, Puebla, México.



Mtra. Erika Torres Sánchez  
UTBIS Puebla Chancellor



Dr. Reyes Juárez Ramírez  
CONISOFT 2022 General Chair





# 2022 IEEE MEXICAN INTERNATIONAL CONFERENCE ON COMPUTER SCIENCE

## CONSTANCIA

A:

*Ivan Y Alba Corpus, Wendy Flores Fuentes, Julio C Rodriguez-  
Quiñonez, Daniel Hernandez-Balbuena, Felix F Gonzalez-Navarro,  
Oleg Sergiyenko, Ruben Alaniz-Plata*

Por su asistencia al **Encuentro Nacional del Computación**, así  
como la presentación de su artículo titulado "*Detection of  
Atypical Data in Point Cloud of Technical Vision System using  
Digital Filtering*" en el **Taller de Inteligencia Artificial: Inteligencia  
Artificial: Métodos, Algoritmos y Aplicaciones**.

Xalapa, Veracruz, México  
24 al 26 de agosto de 2022

**Dra. Karina M. Figueroa Mora**  
Presidenta SMCC

**Dr. Efrén Mezura Montes**  
Organizador General

**Dr. Félix Fernando Navarro González**  
Organizador del Taller



AGOSTO, 24-26

XALAPA, VERACRUZ



# 2022 IEEE MEXICAN INTERNATIONAL CONFERENCE ON COMPUTER SCIENCE

## CONSTANCIA

A:

*Silvia J Urrea Contreas, Brenda L Flores-Rios, María Angélica  
Astorga-Vargas, Jorge Eduardo Ibarra Esquer, Felix F Gonzalez-  
Navarro*

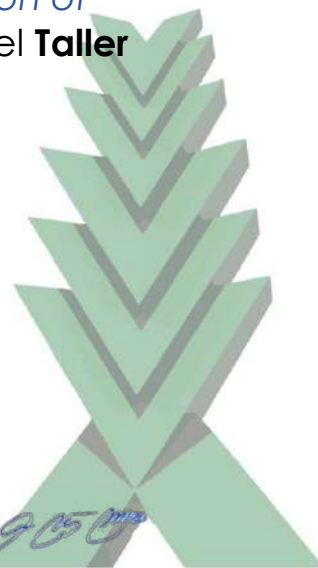
Por su asistencia al **Encuentro Nacional del Computación**, así  
como la presentación de su artículo titulado "*Intervention of  
knowledge flows in software process mining projects*" en el **Taller  
de Ingeniería de Software**.

Xalapa, Veracruz, México  
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**Dra. Karina M. Figueroa Mora**  
Presidenta SMCC

**Dr. Efrén Mezura Montes**  
Organizador General

**Dra. Mirna Ariadna Muñoz Mata**  
Organizadora del Taller



AGOSTO, 24-26

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# 2022 IEEE MEXICAN INTERNATIONAL CONFERENCE ON COMPUTER SCIENCE

## CONSTANCIA

A:

*Luis A Jara Lugo, Jesús Caro, Felix F Gonzalez-Navarro, Mario  
Curiel-Alvarez, Oscar Perez, Nicola Radnev-Nedev*

Por su asistencia al **Encuentro Nacional del Computación**, así  
como la presentación de su artículo titulado "*Nanoparticles  
diameter characterization using image analysis methodology*" en  
el **Taller de Inteligencia Artificial: Inteligencia Artificial: Métodos,  
Algoritmos y Aplicaciones**.

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Organizador General

**Dr. Félix Fernando Navarro González**  
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AGOSTO, 24-26

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*Luis A Jara Lugo, Jesús Caro, Felix F Gonzalez-Navarro, Mario  
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Organizador General

**Dr. Félix Fernando Navarro González**  
Organizador del Taller



AGOSTO, 24-26

XALAPA, VERACRUZ

# Obstacle Coordinates Transformation from TVS Body-Frame to AGV Navigation-Frame

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**Abstract**— This paper describes a mathematical method for obstacle coordinates transformation from a Technical Vision System (TVS) Body-Frame to an Automated Guided Vehicle (AGV) Navigation-Frame. The method is for AGVs equipped with a TVS to scan the scene in front of its travel direction in order to detect the presence of objects and measure their coordinates in the Navigation-Frame to avoid them. The proposed coordinate's transformation method is based on a AGVs kinematic model that includes the calculation of the AGVs load variation influence by the change of the center of gravity of the AGV traveling in an ideal floor conditions.

**Keywords**—automated guided vehicle, navigation, 3D scanning, and technical vision system

## I. Introduction

The AGVs must navigate conflict-free and using optimized paths; should execute their task starting from a predefined initial pose, moving to the desired position. Efficient algorithms are able to avoid obstacles, solve traffic jams and collisions. In the last years, several algorithms for AGV routing have been classified into two categories, static routing, and dynamic routing. In the category of static routing, the route from position  $i$  to position  $j$  is previously determined and then used if the AGV has to be transported from position  $i$  to position  $j$ . Thus, a simple assumption is to choose the route with the shortest distance from position  $i$  to position  $j$ . However, these static algorithms cannot be adapted to changes in the AGV system and traffic conditions. By other side, in the dynamic routing, the information necessary to determine efficient routes are analyzed by intelligent algorithms to make decisions from the various possible routes that can be chosen between position  $i$  and  $j$ , taking into account important features in the routing, such as path maneuvers, time curves, load and unload operations, speed, dispatching rules, and dispatching quantity of AGVs [1], [2]. This paper proposes the use of a technical vision system for obstacles detection on AGVs navigation, and describes a method for obstacle coordinates transformation from TVS body-frame to AGV navigation-frame as information for dynamic routing. The application of the proposed method can be bound in recent works about obstacle avoidance using vision systems such as [3], [4], [5].

## II. AGVs CONFIGURATION, SENSING AND MEASURING

The steering system of an AGV as the function to control the direction of movement; it allows following the desired path. The steering system is generally integrated by linkages, rods, pivots, and gears, some of the systems types are the Ackermann steering, Bell-crank steering, and the Rack and pinion steering, this last one is the most common and composed of a steering wheel, a pinion, a rack & pinion case, and a rack. Some of the most common steering wheel system configurations are skid steer 4 wheels, differential, tricycle wheeled, quad, omnidirectional, and hybrid [6]. Regarding the control architecture which define the behavior control of robots; it is characterized by the way it uses the information collected to analyze time and spatial distribution patterns for decision making, some examples are the deliberative, reactive, behavior based, as well as hybrid systems [7]. By other side the sensors make possible to observe the working environment or guiding the AGV to follow a specified path. Measurements by sensors allow knowing the deviation between the AGV and the path defined in the working environment for further decisions. These sensors can be used on fixed and free path guidance, also in a fusion of them, the most used sensors for the AGVs position and navigation are cameras, laser scanners (reflector-type laser-guided to detect landmarks, triangulation based on reflectors positioned in the environment, dynamic triangulation based on a laser emitter and optoelectronic scanning aperture, Lidar, and laser rangefinder), inertial measurement units, encoders, magnetic sensors, optical sensors systems for measurement base on infrared, signal strength indicators, electromagnetic inductive sensors, and global position systems. With the listed sensors for AGVs have been mentioned position references related with some of those sensors to follow predetermined paths marked with physical and virtual guide-paths. Other sensors are also used for sensing techniques for localization and mapping, without physical guide-paths like a dead reckoning, beacon systems, inertial navigation systems, and computer vision. Some navigation methods provide simultaneous localization and mapping (SLAM), which provide information about the absolute coordinates of the AGV with high accuracy. Others only provide the deviation with respect to a predefined trajectory,



ENC 2021

9-11 agosto  
Morelia, Michoacán



Encuentro Nacional  
de Computación

## CONSTANCIA

A:

*Jesús Albany Armenta García, Félix Fernando González Navarro, Jesús Caro Gutierrez, Brenda Leticia Flores Rios y Jorge Eduardo Ibarra Esquer*

Por su asistencia al *Encuentro Nacional de Computación*, así como la presentación del artículo titulado "*BReML: A Breathing Rate Estimator Using Wi-Fi Channel State Information and Machine Learning*" en el "**Taller de Inteligencia Artificial: Inteligencia Artificial: Métodos, Algoritmos y Aplicaciones**".

Morelia, Michoacán, México  
9-11 de agosto 2021

**Dra. Karina Mariela Figueroa Mora**  
Presidenta SMCC

**Dra. María Lucía Barrón Estrada**  
Organizadora General





ENC 2021

9-11 agosto  
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Encuentro Nacional  
de Computación

## CONSTANCIA

A:

*Emmanuel López Hernández, Felix F. Gonzalez-Navarro,  
Brenda L. Flores-Rios y Jesús Caro-Gutiérrez*

Por su asistencia al *Encuentro Nacional de Computación*, así como la presentación del artículo titulado “*WIFIALR WiFi Alerts Software for Human Movements using Machine Learning Algorithms*” en el “**Taller de Inteligencia Artificial: Inteligencia Artificial: Métodos, Algoritmos y Aplicaciones**”.

Morelia, Michoacán, México  
9-11 de agosto 2021

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Presidenta SMCC

**Dra. María Lucía Barrón Estrada**  
Organizadora General



**La Sociedad Mexicana de Ciencias de la Computación  
y la Universidad Autónoma de Coahuila a través del  
Centro de Investigación en Matemáticas Aplicadas y la Facultad de Sistemas**

otorgan la presente

## **CONSTANCIA**

a

**Marcial Sandoval Gastelum, Félix F. González Navarro and Brenda L.  
Flores Ríos**

Por su destacada participación en el Taller “Inteligencia Artificial: Métodos, Algoritmos y  
Aplicaciones” con el tema “Human Activity Recognition from Wrist Wearable Device  
Sensors using Data Mining Techniques”  
en el Encuentro Nacional en Computación 2020  
efectuado del 24 al 26 de agosto de 2020

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Dra. María Lucía Barrón Estrada  
**Presidenta de la SMCC**

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Dra. Valería Soto Mendoza  
**Organizador General Local**



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a

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Hernández Balbuena and Brenda L. Flores-Ríos**

Por su destacada participación en el Taller “Inteligencia Artificial: Métodos, Algoritmos y Aplicaciones” con el tema “Educational Data Mining for finding patterns in First Semester Engineering Students’ Outcomes to identify potential Success or Failure”  
en el Encuentro Nacional en Computación 2020  
efectuado del 24 al 26 de agosto de 2020

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Dra. María Lucía Barrón Estrada  
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Por su destacada participación en el Taller “Inteligencia Artificial: Métodos, Algoritmos y Aplicaciones” con el tema “Software para Emisión de Alertas basadas en Detección de Movimiento por Señales WiFi”  
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# A Lean Convolutional Neural Network for Vehicle Classification

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**Abstract**—Image classification is an important task in machine vision, in which vehicle classification is used for different applications like traffic analysis, autonomous driving, security, among others. Recent studies made with Convolutional Neural Networks (CNN) have shown that these networks have surpassed older algorithms like Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) in terms of accuracy, speed, and resources management. Even though that CNN have better accuracy and speed they still are heavy in resource consumption on computers which makes them not suitable to deploy on an embedded platform. This paper proposes a lean CNN that has a smaller number of parameters and still maintaining the best accuracy possible on vehicle classification.

**Index Terms**—CNN, Artificial Intelligence, Vehicle Classification,

## I. INTRODUCTION

Former Artificial Neural Networks (ANN) are capable of grouping, pattern recognition, prediction, approximation, among others applications [1] [2], but the arrival of Deep Neural Networks has opened a vast area of applications for artificial intelligence like image classification which can be applied on medical, security, automobile, commerce and other sectors around the world [3] [4] [5].

In recent years the increasing volume of vehicles on city roads has led to the creation of different kind of machine vision applications like security, surveillance, traffic analysis on roads, parking slots, among others [6], where the classification of vehicles is an important task and also is a difficult challenge for image classification algorithms. Vehicle classification has been done with algorithms like K-Nearest Neighbor and Support Vector Machine, the downside of this algorithms is that they are not suitable for remote

applications using hardware with limited resources like an embedded system, due to the characteristics of this methods which is the process of extracting the features in background subtraction of images and also the operations done by the classifier [6], [7], [8], however the arrival of Convolutional Neural Networks which have become a relevant study topic due to their achievements in machine vision applications have turned into the go to algorithms for object classification and even object detection with Regional Convolutional Neural Networks (RCNN). CNN are one of the most used algorithms on image classification and have a wide area of application such as security, health care, traffic monitoring, autonomous driving, among others [9], [10], [11].

Convolutional Neural Networks have obtained great results in object classification which have produced different kinds of distinguished CNN structures like Alexnet, VGG16, GOOGLE-LeNet, which have demonstrated outstanding capabilities in the classification of different kinds of objects [9], [11], [12]. Although these CNN structures have their advantages, they also use a considerable amount of computer resources and are not suitable for embedded systems. Our approach to this kind of complication is to use a lean network architecture proposed for this paper, a CNN that has the minimum number of layers but also has high accuracy, compared with network architecture that requires a significant amount of resources for their training.

CNN are comprised by multiple convolution layers, followed by pooling layers and finally a fully connected Artificial Neural Network (ANN) as seen in Fig. 1. The function of convolutional and pooling layers is to extract the characteristics from the input images to have the best classification possible using the ANN.

In this work we perform a research with CNN to find the

# Classification of Vehicle Images through Deep Neural Networks for Camera View Position Selection

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**Abstract**—This paper shows the experimentation for the classification of vehicle images by artificial vision creating a deep neural network model using Keras and Tensorflow libraries with Python language, a machine learning algorithm. The results are used to select the best camera position to record the vehicular traffic to identify the highway occupancy.

**Keywords**— Artificial vision, deep neural network, Keras, Tensorflow, Python, machine learning, dataset.

## I. INTRODUCTION

Artificial vision is a technology that consists of methods for the acquisition, processing, and analysis of data through images for the creation of usually numerical data sets so that they can be processed using computer algorithms [1], [2], [3], [4].

Deep neural networks analyze numerical and characteristic data. They extract characteristics and classify patterns.

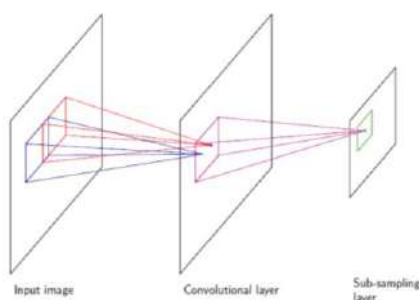


Figure 1. Diagram illustrates the part of a convolutional neural network.

These algorithms have been used to find patterns of economic fraud, speech recognition, signals, and images, make predictions in the financial market, make weather forecasts, etc.

Keras is a neural network library written in the Python programming language. It can run on Tensorflow. It is designed to quickly experiment in Deep Learning Networks. Tensorflow is a library to build and create neural networks for Machine Learning.

Machine Learning is the algorithm that can learn on its own through datasets, that is, to recognize patterns of this data [5], [6], [7].

A dataset is a collection of data that is mostly tabulated. A dataset corresponds to the contents of a single database table [8]. The Stanford University dataset was chosen because it presents cars of normal angles (Front, rear, side, and tilt) in which one could have images of these as seen in Figure 2.

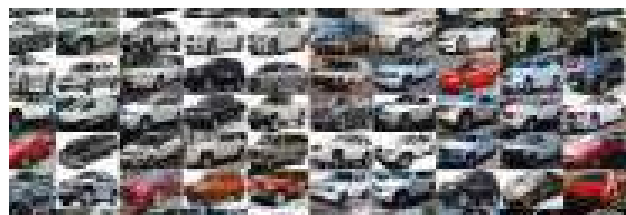


Figure 2. Dataset from Stanford University.

As the deep neural network must learn alone to recognize a wide variety of objects within an image, then we will have a very large dataset, the more images we have for training, the better it will capture its characteristics, that is, it will better classify the objects.

There is a study of Bilinear CNN Models for Fine-grained Visual Recognition which uses the Stanford University dataset. This study proposes “bilinear models, a recognition architecture that consists of two feature extractors whose outputs are multiplied using the outer product at each location of the image and pooled to obtain an image descriptor. This architecture can model local pairwise feature interactions in a translationally invariant manner which is particularly useful for fine-grained categorization. The dataset used is that of Stanford University”, [9].

Generic image classification uses auto-bilinear grouping to extract image characteristics. Discrimination of several categories is recommended with the extraction of finer characteristics at the same time.

A paired bilinear grouping operation between the base and query images is proposed to extract fine-grained features. Meanwhile, their relationships are explored by a



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# An MHD Stirrer 2D Velocity Profile Measurement Validation Through a Machine Vision System

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**Abstract**— The paper presents a technique developed for the enhancement of a Particle Image Velocimetry (PIV) measurement process in a customized machine vision system. The PIV measurements are performed for an accurate “Velocity Profile” mathematical model validation of a Magnetohydrodynamic (MHD) stirrer prototype. Data mining and filtering have been applied to a raw measurement database from the customized machine vision system designed to evaluate the MHD stirrer prototype. Outlier’s elimination and smoothing have been applied to raw data to approximate the PIV measurements output to the velocity profile mathematical model in order to increase the accuracy of a customized machine vision system for 2D velocity profile measurements. The accurate measurement of the 2D velocity profile is fundamental owing to the requirement of future enhancement of the customized machine vision system to construct the 3D velocity profile of the MHD stirrer prototype. The presented technique can be used for measurement and validation in the design of devices that require fluid manipulation for tasks such as pumping, networking, propelling, stirring, mixing, and even cooling without a need for mechanical components due to MHD’s non-intrusive nature provides a solution to mechanical moving elements issues.

**Keywords**— data mining, filtering, magnetohydrodynamics, particle image velocimetry, measurements, machine vision.

## I. INTRODUCTION

The need for measuring the velocity of flowing stream in fluidic channels has given the origin to the particle image velocimetry technique (PIV). In the beginning, the first velocity flow measurements were done tracking the trajectory of debris over the flow surfaces of natural channels, such as rivers. With the development of the mechanics of fluids and the innovation in the use of fluids and microfluidics for a lot of applications, it was necessary to seed particles in the fluid and developed a technique to track their trajectory, measure their velocity and define its moving patterns, obtaining an accurate, quantitative measurement of fluid velocity vectors at a very large number of points simultaneously [1]. Besides due to the

development of computers, sophisticated machine vision systems for PIV have been designed for different applications. Where the PIV technique challenge and complexity depend on the characteristics of the application, as well as in the required resolution and accuracy. Examples of systems that use the PIV techniques to characterize its performance are the turbulent jets used in a wide variety of applications, as in marine propulsion, pollutant discharge or studies on the cardiocirculatory system [2]. The turbines for hydropower purposes to analyze from complex turbulent to two-phase flows [3]. Centrifugal pumps to measure the complex flow structures around the volute tongue region, where intense rotor-stator interaction appears [4]. Solid-liquid laminar stirrer tank with floating particles for industrial processes, such as the production processes of coatings, cosmetics and lubricating oil [5]. And microfluidics in annular channels for MHD stirrers [6]. Since micro-applications started to develop and surged the need of propelling, stir, and control flows to perform tasks such as mixing and networking at micro-scale, prototypes based on the principle of Magnetohydrodynamics (MHD) started to be considered [7]. MHD prototypes enable fluid manipulation for tasks such as pumping, networking, propelling, stirring, mixing, and even cooling without the need for mechanical components, its non-intrusive nature provides a solution to mechanical systems issues. The lack of mechanical elements reduces the possibilities of breakdowns, undesired vibrations, and noise. MHD is particularly useful for microfluidic applications where a conductive fluid needs to be manipulated to perform specific operations. The most common applications of MHD are micro-pumping (in lab on chip (LOC’s) and liquid chromatography (LC)) [8], [9], [10], networking (in LOC’s) [8], stirring and mixing (in LOC’s, polymerase chain reaction (PCR), and chemical reactors) [11], [12], [13], micro-cooling (in microelectronics), and thermal cyclers [14], [15], [16]. However, the design of MHD prototypes is not an easy task. The design requires a precise flow control, which depends on several parameters, like microfluidics conductivity, microfluidics channel shape and size configuration, and the



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# CARACTERIZACIÓN DE LA SATISFACCIÓN DEL CLIENTE Y CALIDAD DEL SOFTWARE EN PYMES DE BAJA CALIFORNIA, MEXICO

M.T.I.C. Roberto Carlos Valdés Hernández<sup>1</sup>, Dr. José Luis Arcos Vega<sup>2</sup>, Dr. Félix Fernando González Navarro<sup>3</sup>

**Resumen-** Actualmente el ámbito de desarrollo de software ha alcanzado un desarrollo impensable en el mundo, las empresas de este sector con el objetivo de seguir compitiendo buscan sustentar la calidad del software y la satisfacción del cliente. En esta investigación se presentan los resultados sobre la caracterización en las dimensiones de satisfacción del cliente y calidad del software de las pequeñas y medianas empresas (Pymes), del Estado de Baja California; con una participación de 52 empresas, los resultados obtenidos se analizaron realizando un análisis de correlación en las variables calidad y satisfacción del cliente utilizadas en el instrumento, el cual se formuló con 31 reactivos aplicándose a los directivos de las empresas. Los resultados muestran que tanto la satisfacción del cliente y la calidad tienen una influencia positiva y significativa en el proceso de desarrollo de software en las Pymes.

**Palabras Claves** – Pymes, Competitividad, Satisfacción del cliente, Calidad y Procesos de software.

## Introducción

En la actualidad, las empresas buscan mantenerse en un mundo cada vez más competitivo, y en un marco económico caracterizado por la globalización e internacionalización de los mercados, en un ambiente dinámico producido por cambios continuos que impactan en la gestión de las empresas, necesitando establecer esquemas integrales de gestión, buscando con esto mejorar los procesos (Escobar, Moreno, & Cuevas, 2016). En México según Ortiz & Arredondo, (2014) las MIPyME representan 95% del total de las empresas, al tener en mente empresas de pequeña magnitud, frecuentemente se considera que esto se debe a su incapacidad para desarrollarse y las coloca en desventaja frente a las grandes empresas. Según la CEPAL (2010) durante el primer año de vida de las empresas aproximadamente 50% mueren, durante el segundo año otro 25% y antes de llegar al quinto año otro 15%, sobreviviendo y madurando como pequeña empresa 10% de las empresas inicialmente creadas. En este sentido, en las empresas dedicadas a desarrollar software y de otro giro, existen roles de trabajo que están asignados a profesionales capaces de llevar a cabo las tareas asignadas, mediante la aplicación de metodologías de desarrollo en procesos de producción de software, siendo uno de los más complejos en empresas de este sector, en donde la asignación de los roles para dicha actividad tiene un grado alto de dificultad, ya que para la asignación de dichos roles se deben de considerar las habilidades y conocimientos requeridos a distintos niveles en el proceso de desarrollo de software, siendo esta una actividad compleja que involucra un conjunto de fases altamente creativas, en la que las actividades forman parte del propio proceso productivo, enfocando las acciones de las pymes en hacer eficientes sus procesos, esto como parte de su estrategia de negocio, tanto en términos estrictamente técnicos, como organizacionales y comerciales. Es importante reflexionar que, en el proceso de desarrollo de software, la utilización de técnicas y modelos por parte de las pymes son un pilar fundamental, ya que estos están orientadas a asegurar la calidad del software (Rojas, Pino, y Martínez, 2015). Las pymes han creado su propia línea de trabajo Strub (2015) en cuanto a la gestión de la calidad se refiere, es decir, trabajan sobre el proceso de producción de software para proveer calidad en sus productos y así satisfacer las necesidades de sus clientes, por lo que el utilizar certificaciones las cuales permitan testificar la eficacia y eficiencia de los procesos, buscando mejorar las formas de alcanzar los objetivos (Esterkin, & Pons, 2017). En este sentido, la calidad de software según Fenton & Bieman (2014) se puede medir en términos de densidad de defectos, por lo que las empresas dedicadas a desarrollar software Puello et al. (2016) deben aplicar métodos de control los cuales permitan que el producto desarrollado cumpla con las normas y estándares de calidad, los cuales obedezcan a las expectativas del cliente. Considerando lo anterior, es importante asegurar la calidad del producto software haciendo énfasis al talento humano Chavarría, Oré & Pastor (2016) considerando que las personas deben de comprometerse con la calidad del producto a desarrollar, siendo este uno de los componentes que hoy en día está jugando un papel fundamental en la industria y es un factor determinante en la satisfacción del cliente (Holguín, 2014).

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## FW: W3PHIAI-19 notification for paper 12

1 message

成井 公一 / Hirokazu Narui <hirokazu.narui@furukawaelectric.com>

Wed, Nov 28, 2018 at 1:34 PM

To: Stefano Ermon <ermon@cs.stanford.edu>, Rui Shu <ruishu@stanford.edu>, "fernando.gonzalez@uabc.edu.mx" <fernando.gonzalez@uabc.edu.mx>

Cc: 横内 則之 / Noriyuki Yokouchi <noriyuki.yokouchi@furukawaelectric.com>

Dear Stefano, Rui, Fernando

My paper "Domain Adaptation for Human Fall Detection Using WiFi Channel State Information" is accepted for a poster presentation at AAAI workshop (W3PHIAI-19).

I need to submit the camera-ready poster paper (up to 4 pages) until Dec 16th.  
It will be published workshop proceedings as a Springer book.

Would you check the draft before submission?  
I'll prepare it by Dec 11th...

Thank you for giving a lot of advices while I was at Stanford.  
I'm really appreciate for collaborating with me!

Best,  
Hiro

\*\*\*\*\*

Hirokazu Narui / [hirokazu.narui@furukawaelectric.com](mailto:hirokazu.narui@furukawaelectric.com)

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2018/11/28 6:32 に、"w3phiai19@easychair.org (W3PHIAI-19 の代理)" <w3phiai19@easychair.org> を書き込みました:

Dear Hirokazu Narui,  
Congratulations! Your paper titled 'Domain Adaptation for Human Fall Detection Using WiFi Channel State Information' (Submission 12) has been accepted for a poster presentation at the W3PHIAI-19 workshop to be held in conjunction with the 33rd AAAI Conference on Artificial Intelligence (AAAI-19) on Jan 27th and 28th, 2019 at the Hilton Hawaiian Village in Honolulu, Hawaii. We received a record number of submissions this year and your paper was reviewed by a group of experts and selected for inclusion by the workshop organizing committee. Please read this email in its entirety for information about your presentation, the workshop, and the camera-ready version of your manuscript.

Posters will be presented during the workshop's poster session to be held on Sunday, January 27th from 1:30 pm to

# Domain Adaptation for Human Fall Detection Using WiFi Channel State Information

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## Abstract

We develop a novel unintrusive system for human fall detection combining WiFi Channel State Information (CSI) and machine learning. We only use a WiFi transmitter and receiver, without requiring the user to carry any sensors. The CSI is affected by motions in the environment, and human activities can in principle be detected using machine learning. However, the CSI varies from one environment to another, requiring the collection of environment-specific training data. To overcome this challenge, we propose a combination of convolutional neural network and domain adaptation. By adapting to un-labeled data from a new environment, we significantly improve precision and recall, making deployment possible to new, previously unseen environments.

## 1 Introduction

Human fall detection plays a tremendous role for elderly care in indoor environments. From the (CDC 2013), one out of five falls causes serious injury such as broken bones or a head injury. At the worst case, elderly person may not move or lose consciousness after an accident and there might be no chance for others to be notified about it if the person has been alone. Due to such high number of seniors living alone, the human fall detection system has become a priority for the health care industry.

Human fall detection techniques can be categorized into device-based or device-free approaches. Device-based systems use motion sensors such as a gyroscope or an accelerometer implemented in smart devices (Jiangpeng et al. 2010) or wearable sensor (Bianchi et al. 2010). However, it is inconvenient for people to carry or wear a device all the time at home, and they might need to take it off especially while taking a bath or when it needs to be charged. On the other hand, a device-free system does not require a wearable device. Some existing systems are based on camera and image sequence-based action recognition (Homa et al. 2008; Homa, Aski and Pourreza 2008). These system can release the inconveniences of carrying or wearing smart devices for detecting "Fall". Although camerabased systems' high accuracy in activity recognition come at the cost of the user's

privacy in their own home. Moreover, the camera based approach has fundamental limitations of working only in line-of-sight (LOS) situation and when there is sufficient lighting present.

To overcome these limitations, other systems are based on wireless technologies such WiFi access points (Yuxi, Kaishun, and Lionel 2017), and radars (Dorp and Groen 2008). WiFi-based approaches have the advantage of being more cost efficient than radars. Furthermore, WiFi systems are already ubiquitous in homes and buildings. As a result, WiFi CSI has been studied for use in several applications such as human activity recognition (Chowdhury 2018, Wang et al. 2015), recognizing keystrokes (Ali et al. 2017), indoor localization (Vasisht, Kumar, and Katabi 2016; Kotaru et al. 2015), human gesture recognition (Pu et al. 2013), crowd counting (Xi et al. 2014) and human identification (Zhang et al. 2016) by extracting corresponding features.

While many of these techniques work well in the specific environment they were designed for, the performance of WiFi CSI-based systems can degrade in other, significantly different environments. We have observed that this is caused by covariate shift (Shimodaira 2000) between training and test datasets. The main contribution of this paper is hence two-fold:

- To overcome the covariate shift caused by changing environments while using WiFi CSI, we propose a 1-dimensional convolutional neural network (CNN) model with domain adaptation. We evaluate WiFall and CNN model with and without domain adaptation (VADA and DANN) using our WiFi Activity dataset.
- Using T-SNE, We extensively analyze how CSI feature distribution varies across different environments. We also compare the feature distribution between a 1-dimensional and 2-dimensional CNN.

Using the dataset we collected and tagged for several different activities, we have compared precision and recall between a conventional machine learning techniques and some deep learning approaches with and without domain adaptation. The proposed technique with domain adaptation shows higher precision and recall than the other ones. Using T-SNE, we confirm that domain adaptation works well for



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# 2018 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops)



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# A Lifelogging Platform Towards Detecting Negative Emotions in Everyday Life using Wearable Devices

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**Abstract**— Repeated experiences of negative emotions, such as stress, anger or anxiety, can have long-term consequences for health. These episodes of negative emotion can be associated with inflammatory changes in the body, which are clinically relevant for the development of disease in the long-term. However, the development of effective coping strategies can mediate this causal chain. The proliferation of ubiquitous and unobtrusive sensor technology supports an increased awareness of those physiological states associated with negative emotion and supports the development of effective coping strategies. Smartphone and wearable devices utilise multiple on-board sensors that are capable of capturing daily behaviours in a permanent and comprehensive manner, which can be used as the basis for self-reflection and insight. However, there are a number of inherent challenges in this application, including unobtrusive monitoring, data processing, and analysis. This paper posits a mobile lifelogging platform that utilises wearable technology to monitor and classify levels of stress. A pilot study has been undertaken with six participants, who completed up to ten days of data collection. During this time, they wore a wearable device on the wrist during waking hours to collect instances of heart rate (HR) and Galvanic Skin Resistance (GSR). Preliminary data analysis was undertaken using three supervised machine learning algorithms: Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA) and Decision Tree (DT). An accuracy of 70% was achieved using the Decision Tree algorithm.

**Keywords**— *Wearable Devices; Pervasive Computing; Lifelogging; Emotion Recognition; Classification*

## I. INTRODUCTION

Negative emotions, such as stress and anxiety, are a common occurrence in daily life and can be seen as a characteristic of a reaction to certain situations or triggers [1]. However, frequent and repeated episodes of stress have been associated with the process of inflammation in the human body [2]. Furthermore, the frequency and magnitude of stress that are experienced by the individual may have a cumulative effect on the cardiovascular health [3]. There is evidence that repeated and sustained activation of the sympathetic nervous system, alongside other physiological changes, such as increased adrenaline, cortisol, and levels of proinflammatory cytokines, is associated with age-related diseases, such as cardiovascular disease, diabetes, arthritis [2]. Given that between 9% - 11% of adults in the UK suffer from cardiovascular disease (CVD) [4], those self-regulatory processes underpinning the recognition/reduction of stress assume increased significance with respect to public health.

It is envisioned that lifelogging technologies can be used as a platform to develop self-awareness in order to support coping strategies that reduce the physiological impact of stress. Lifelogging is a form of pervasive computing and can be described as, “a unified digital record of the totality of an individual’s experiences, captured multimodally through digital sensors and stored permanently as a personal multimedia archive” [5]. This paradigm allows individuals to automatically collect vast amounts of personal data over their lifetime. These data can then be used to promote self-reflection, self-experimentation and to support behaviour change. However, in order to avoid the burden of excessive data entry, it is important for lifelogging platforms to record data *automatically* and *unobtrusively*. Modern smartphones and wearable devices contain a number of non-intrusive sensors that enable collection of an enormous amount of personal information. The feasibility of these mobile technologies for lifelogging is supported by its increasing affordability and pervasiveness, i.e. sensors are becoming cheaper and smaller. Additionally, smartphones sit within a growing category of networked devices that can be used to monitor the behaviour of the individual [6].

This paper presents a mobile lifelogging platform that has been developed to: (1) collect instances of data, including heart rate (HR) and Galvanic Skin Resistance (GSR) from a wearable device, (2) synchronise and remove noise and artefacts from the collected data, and (3) apply machine learning techniques to classifying instances of stress. Physiological markers of stress are monitored through the collection of psychophysiological data (HR and GSR). The usage of implicit monitoring delivers quantitative data during everyday life, which can be transformed into visualisations that support accurate recall from long-term memory.

The remainder of this paper is constructed as follows. Section 2 provides an overview of background literature, whilst section 3 illustrates the design of the posited system. Section 4 describes the pilot study that was undertaken to collect data in everyday life from undergraduate university students using wearable devices. This section also presents subsequent signal pre-processing and labelling that was undertaken on this data. Section 5 presents results of preliminary analysis to classify stress using the data described in section 4, whilst section 6 presents a discussion of the results. The paper is then concluded in section 7.

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## ACADEMIA JOURNALS



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# LA SATISFACCIÓN DEL CLIENTE EN LA UTILIZACIÓN Y CALIDAD DEL SOFTWARE DE LAS PEQUEÑAS Y MEDIANAS EMPRESAS (PYMES) EN BAJA CALIFORNIA

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Dra. Brenda Leticia Flores Ríos<sup>4</sup>

**Resumen-** En este estudio se presentan los resultados de las pequeñas y medianas empresas (PYMES), dedicadas al desarrollo de software en el Estado de Baja California; considerando como objetivo principal el análisis sobre la satisfacción del cliente y la calidad del software, dos factores claves de esta industria. La muestra de este estudio son 52 PYMES dedicadas al desarrollo del software y que tienen un número de empleados entre 5 y 250, para el cual se diseñó un instrumento que consta de 32 preguntas y su aplicación fue por vía electrónica. Los resultados obtenidos muestran como las empresas gestionan los proyectos de desarrollo con mayor eficiencia, ya que los clientes son una parte fundamental para conocer la calidad de los productos.

**Palabras Claves** – Pyme, Clúster, Satisfacción del cliente y Calidad.

## Introducción

Dado el avance de la Ingeniería de Software en la actualidad, se vive una etapa en la que el software ha alcanzado un desarrollo impensable, la industria de este sector ha evolucionado con el paso de los años, con el objetivo de seguir compitiendo, en donde parte de esa competitividad se da a través de sustentar la calidad del software y la satisfacción del cliente. Según Dapozo *et al.*, (2015) los métodos y herramientas son un pilar fundamental ya que están orientados a mejorar la calidad del software, considerando que los atributos de calidad del software es una necesidad para las empresas de software, ya que esto permite mejorar las tareas menos eficientes y obtener productos de calidad, como parte fundamental de su estrategia empresarial. Esto se traduce en una ventaja competitiva para las empresas, ya que la medición permite modificar aquellos factores que aportan una mayor eficacia en el proceso productivo y contribuyen a la calidad del producto (Hernández y Minguet, 2005).

Considerando lo anterior, diversos autores (Rogers, 2005; McCallhan, 1977) tratado de determinar qué factores son los que afectan la calidad del software coincidiendo que esta se da a través de una combinación de factores, para lo cual se necesita medir los atributos que afectan dicha calidad, y estos pueden ser medidos durante el proceso de desarrollo, lo cual lleva a la empresa y específicamente al encargado del proceso a tener una aproximación en tiempo real de la eficacia del análisis, diseño y codificación, y de la calidad del software que se desarrolla. Dentro de la industria del software, las Pymes dedicadas a este rubro reconocen la importancia de entregar a sus clientes un producto de software de calidad, ya que es un pilar importante en el proceso de desarrollo, en un mercado que exige productos de alta calidad, el cual satisfaga sus necesidades. (Piattini *et al.*, 2014). Es importante mencionar que las Pymes utilizan técnicas y modelos en el proceso específico, a través de las cuales se definen las actividades y técnicas de pruebas utilizadas en el proceso de desarrollo de un producto software, buscando con esto asegurar la calidad del software (Rojas, Pino, y Martínez, 2015). En este sentido, como sociedad somos testigos de cómo hemos ido evolucionando, una sociedad en donde la calidad tiene una relevancia importante en todo el entorno económico. Para las empresas de software, un factor crucial en el proyecto de desarrollo son los métodos a utilizar en dicho proceso, orientados a mejorar la calidad del software, en donde una estimación deficiente del proceso y duración del proyecto, puede ocasionar pérdida de competitividad e incumplimiento de entrega del producto (Gómez *et al.*, 2014). Por lo anterior, es indispensable dar seguimiento continuo del producto durante su ejecución, mejorando la eficacia de los factores durante el proceso productivo y contribuyendo así a la calidad del producto, en donde varios países como Malasia, Filipinas o Tailandia

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*Roberto-Carlos Valdés Hernández, José-Luis Arcos Vega y  
Félix-Fernando González-Navarro*

*por el artículo titulado*

**Análisis del Desarrollo del Software en Pequeñas y Medianas Empresas (PYMES)  
del Estado de Baja California, México**



**Nagib C. Callaos**  
Presidente del Comité de Programa  
CICIC 2017

24 de marzo de 2017  
Orlando, Florida, EE.UU.



**Belkis Sánchez**  
Presidente de la Comisión Organizadora  
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## **Análisis del desarrollo del Software en Pequeñas y Medianas Empresas (PYMES) del Estado de Baja California, México.**

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### **RESUMEN**

Este documento presenta un estudio realizado sobre el desarrollo de software en pequeñas y mediana empresas (PYMES) dedicadas a esta actividad del estado de Baja California, con el propósito de generar información sobre tres dimensiones, el desarrollo, satisfacción del cliente y calidad, considerados como factores importantes en la industria del software de la región. Las Pymes que conforman la muestra de estudio, son organizaciones que cuentan con un número de empleados entre los 5 y 250, y que se encuentra en condiciones de crecimiento. Durante el trabajo de análisis, se realizaron encuestas vía correo electrónico a los gerentes de las PYMES, así como entrevistas personales. El análisis de la información reveló que cada una de las dimensiones antes mencionadas impacta de manera directa en el proceso de desarrollo del software, así como información percibida por el cliente en cuanto a la satisfacción, y la calidad con la que se realiza el producto de software ofrecidos por estas empresas. Asimismo, los resultados obtenidos corresponden a 52 PYMES dedicadas al desarrollo de software, lo que permitió identificar la relación que existe entre el desarrollo, satisfacción del cliente y la calidad, y cómo influyen en los resultados de esta industria, buscando contribuir a la mejora continua del software, ya que dentro de las actividades de las PYMES involucradas en este estudio, que están encaminadas a la satisfacción del cliente y que involucran a varios actores externos que los exhortan a participar en las actividades de mejora continua e innovación y la importancia de anticiparse a las necesidades potenciales y futuras del cliente, en donde la calidad de los procesos juega un papel importante en el desarrollo del software. Cabe mencionar que, en los resultados obtenidos, las comparaciones realizadas entre reactivos aplicados en el instrumento de evaluación, bajo la perspectiva de las tres dimensiones requeridas para el

desarrollo de software de las PYMES, mismos que sirvieron para evaluar e identificar las percepciones de las personas encargadas de desarrollar estos procesos.

Palabras Claves: Estructura organizacional, Software, Desarrollo, PYMES, Satisfacción del cliente y Calidad.

### **1. INTRODUCCIÓN**

Las empresas dedicadas al desarrollo de software considerando este como el conjunto de instrucciones realizadas por un programador o varios programadores para que una computadora realice una o varias acciones, a través de las cuales se procesan, almacenan y comunican un contenido informativo [1], en donde estas empresas fueron aprobadas como PYMES en México en el año 2002, en donde la secretaria de economía reconoce que el proceso de adopción de modelos y estándares es costoso, lo cual es imprescindible para cumplir con lo que establece el Reglamento de la Ley de Adquisiciones, Arrendamientos y Servicios del Sector Público deben ser lo suficientemente accesibles en términos de costo, a fin de que no sean una barrera de entrada para las empresas pequeñas y medianas, cuya clasificación publicada en el diario oficial de la federación del 2002 [2], en donde se describe la clasificación como empresa pequeña de 31 a 100 empleados, mediana, de 101 a 500, pequeña de 11 a 50, y mediana de 51 a 250.

El software, las telecomunicaciones y la informática constituyen los pilares de la nueva industria, ya que proveen a otras industrias y a la vez demandantes de otros bienes y servicios de muchas más, y esta nueva industria se amplía o desarrolla continuamente a la par de las tecnologías de la información [14]. En México, el tamaño del mercado de las TIC pasó de 18 mil millones en 2001 a 33 mil millones en 2008, poco menos de la mitad del de





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Popayán, Colombia, Octubre 2 de 2016

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**ASUNTO: Participación Autor: “11 Congreso Colombiano de Computación”**

Un Cordial Saludo,

La presente tiene como objetivo informar de su participación del **11vo Congreso Colombiano de Computación (11CCC)** realizado en la ciudad de Popayán, Cauca - Colombia del 27 al 30 de Septiembre del 2016. Durante el evento ustedes participaron como autores del trabajo 109 titulado “Process Mining in Software Process Improvement, a Systematic Literature Review”, el cual fue presentado el 29 de Septiembre en la línea de Ingeniería de Software con el Chair PhD. Julio Ariel Hurtado, en las instalaciones de la Casa de la Moneda, en esta ciudad.

El Congreso Colombiano de Computación es el encuentro informático anual organizado por la Sociedad Colombiana de Computación. Tiene como objetivo propiciar el fortalecimiento de la comunidad colombiana en informática, acercando a investigadores, estudiantes y practicantes, tanto nacionales como internacionales. En la edición del 11CCC, se contó con la participación de reconocidos investigadores de 13 países en el área de la Computación e informática y estuvo organizado por la Sociedad Colombiana de Computación, por la Universidad del Cauca, la Fundación Universitaria de Popayán, la Alcaldía de Popayán y por IEEE Colombia.

Agradeciendo su colaboración,

Atentamente,

Phd. César Alberto Collazos  
Presidente Comité Académico 11CCC

# *Process Mining in Software Process Improvement, a Systematic Literature Review*

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**Abstract**— In this review, we observed there is a growing interest on the strategy of applying the discipline of Process Mining in Software Process Improvement projects. Process Mining aims to discover, monitor and improve processes through analysis of the various event logs generated by organizational processes. In order to identify the predominant contributions, the results were classified in three categories: Theoretical Foundations and Algorithms, Proposals and Tools. We found emerging methodologies and tools with characteristics that enable the application of this discipline into Software Improvement projects.

**Keywords**—*Process Mining, Software Process Improvement Project, Systematic Literature Review.*

## I. INTRODUCTION

Software Process Improvement (SPI) has become the most recommended discipline to drive the increase of business competitiveness [1]. The successful implementation of SPI has provided important benefits to software organizations; increase in customer satisfaction, efficiency in development process, definition and accomplishment of quality objectives among others [2].

In an SPI project, the assessment process establishes that information must be compiled, registered, stored, processed, analyzed and presented via tools based on paper or software. Mainly, software tools are a valuable support for evaluators allowing them to compile, register and classify evidence especially when volume and information complexity are significant [3]. However, the amount of information increases for each process or improvement cycle, causing scalability or evidence control problems. In general, size of knowledge databases or repositories of processes execution logs are based on criteria of capacity and efficiency of storage and not on the future usage, exploration, exploitation or analysis of those logs. Hence the limited access to such information prevents the enrichment of criteria for decision making needed in the discovery of new business rules, continuous improvement, monitoring and control of the already established processes [4].

This document presents by means of a Systematic Literature Review (SLR), how Process Mining (PM) can provide techniques, algorithms or tools when an SPI project has already generated a considerable volume of information.

Thus, deployed processes and information would be perfectly aligned to the related requirements with norm compliance, efficiency and customer support [5].

This document is structured as follows: In section two, we propose a conceptual framework with information regarding Data Mining, PM and SPI; Section three describes the research methodology; to later address, in section four, the results on the interrelationship of PM with the improvement of software processes based on CMMI-DEV. Finally, conclusions and future work are presented.

## II. STATE F ART

Data Mining and PM are common disciplines that provide the techniques and tools for such analysis, making use of algorithms that search for patterns and relationships. For this reason, it is important to present the formal definition and properties of both disciplines.

The Statistical Analysis Systems Institute defines the term Data Mining as the process of selecting, exploring, modifying, modeling and valuating huge amounts of data with the goal of discovering unknown patterns which can be used as a competitive advantage for an organization [6]. Nowadays, it is used to refer to a set of global processes for knowledge discovery from data and data analysis algorithms.

Table I shows definitions of PM found in literature. In general, it is specified as a discipline which aims to discover, monitor, and improve processes through the extraction of knowledge from event logs in information systems [6]. Event logs relate to the execution of organizational processes history, where instances or process cases are found. Case studies were found about the deployment of PM in hospital processes [4], industrial systems [7], government dependencies, biology, medicine, economy, among others. These are available in Process Aware Information Systems (PAIS), as are in workflows, BPMS, ERP or CRM [6].

According to Aguirre Mayorga [8], PM provides tools (ProM, CPN, DisCo) and techniques that help organizations to: 1) Discover the real process execution model; 2) Determine if the process complies with the regulation and documented procedures; 3) Analyze the interaction of staff running the process; 4) Discover bottlenecks; 5) Predict the cycle time of a

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- Ballroom 2
- Ballroom 3
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- Ballroom 5
- Copper
- Gold
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# OCR for Unreadable Damaged Characters on PCBs Using GSC Algorithm and kNN Classifier

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**Abstract**— In this paper, we propose to change the actual implemented pattern matching method to have optical character recognition by implementing the Gradient, Structural, Concavity (GSC) algorithm to extract the features of damaged, unreadable or incomplete numerical digit characters from images on printed board circuits (PCBs). Grayscale color images are acquired from a charge-coupled device (CCD) camera, assembling a dataset of 500 matrix images samples for the character digits from 0 to 9. The GSC feature extraction method is applied to get the characteristics that will be used in the character recognition step. Experimental results show that applying GSC algorithm to extract the features and using k-Nearest Neighbor (kNN) Classifier with the Euclidian Distance can improve optical character recognition (OCR) detectability of damaged characters from actual 95% to more than 97% in early tests.

**Keywords**— *machine vision, optical character recognition (OCR), kNN Classifier, GSC, pattern recognition, printed circuit board (PCB).*

## I. INTRODUCTION

Optical character recognition (OCR) has been an important technology used to convert characters from a digital image to a digital text. There are basically two types of OCR algorithms: the first technique is related with the matching of matrix images, where an alphabet of stored character images is used to compare with an input image [1], [2]. This pattern matching does not work well when new fonts are encountered or input character images are unreadable. The second technique decomposes an input image to extract the principal features [3], [4], [5]. Then, classifiers are used to compare the input image features with some stored image features and choose the best match.

Our actual system implemented at the Skyworks factory uses the traditional OCR technique i.e. pattern matching. Our implemented vision system reads identification characters on printed circuits boards (PCBs) for lot integrity and machine control. This commonly-used technique is not robust enough because many of the images on PCBs present some damaged characters due to dirt or as a result of bad previous processes [1]. Actual OCR detectability is around 95% at best. It starts with a monochrome VGA image acquisition of the upper left section of a PCB, using a NI-1752 smart camera, with full resolution, 640x480 pixels and maximum data transfer @60 fps using a GigE port. The selected resolution and data transfer speed parameters meet the factory production schedule of inspected PCBs. The camera has a grayscale output image type

with a maximum character resolution to cover the entire PCB characters positions, as shown in Fig. 1.



Fig. 1. PCB with no damaged characters.

Due to some problems with previous processes in the production line, some PCBs will have some residual dirt over the characters, making some characters unreadable for the pattern matching technique, as shown in the following Fig. 2.



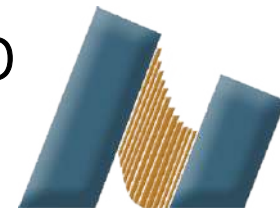
Fig. 2. PCBs with evident residual dirt over characters.

The principal problem is that operators have lower throughput than automatic OCR software, and this leads to manually writing down the information from the screen when the actual recognition software fails, increasing the process time, making possible errors from wrong readings, resulting in higher production costs. Taking into consideration these facts, a better approach has to be considered [3].

This paper presents a proposal for implementing a character recognition technique for unreadable characters using Gradient, Structural and Concavity (GSC) extraction features and K-Nearest Neighbor Classifier using Euclidian Distance [6], [7], [8].



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Por su participación en la Sesión de Posters dentro del II Simposio de Nanociencias y Nanomateriales.  
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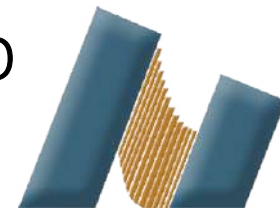


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# OCR for Unreadable Damaged Characters on PCBs Using Principal Component Analysis and Bayesian Discriminant Functions

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**Abstract**— In the last few decades, new computer vision technologies and image processing techniques have been very important in the improvement and automation of manual processes in many technical areas, e.g., in the semiconductor industry. In this paper, we propose to change the actual pattern matching methods implemented to have optical character recognition by the use of principal component analysis method to extract the principal characteristics and features of damaged or unreadable numerical digit characters from images on printed board circuits (PCBs) and compute linear and quadratic Bayesian discriminant functions to classify and find the correct numerical character that corresponds to those features. In the first step of this work, grayscale color images are acquired from a charge-coupled device (CCD) camera, then image segmentation is manually computed to create a dataset of 500 matrix images for the character digits from 0 to 9. Then, a feature extraction method is applied to get the principal components that will be used in the character recognition state. Finally, our results show that applying Bayesian linear and quadratic discriminants to the principal component features can improve optical character recognition (OCR) detectability of damaged characters from actual 95–97% to 99.88% in early tests. This suggests to us that the problem probably follows a linear model where linear hyperplanes separate decision regions with satisfactory (almost no) errors.

**Keywords**— *machine vision; optical character recognition; Bayesian discriminant functions; principal component analysis; linear discriminant classifier; quadratic discriminant classifier.*

## I. INTRODUCTION

Optical character recognition (OCR) has been an important technology used to convert characters from a digital image to a digital text. There are basically two types of OCR algorithms: the first technique is related with the matching of matrix images, where an alphabet of stored character images is used to compare with an input image [1], [2]. This pattern matching does not work well when new fonts are encountered or input character images are unreadable. The second technique decomposes an input image to extract the principal features [3], [4], [5]. Then, classifiers are used to compare the input image features with some stored image features and choose the best match.

Our actual system implemented at the Skyworks factory uses the traditional OCR technique, pattern matching. Our

implemented vision system reads identification characters on printed circuits boards (PCBs) for lot integrity and machine control. This commonly used technique is not robust enough because many of the images of PCBs shown some damage on the characters due dirt or the results of bad previous processes [1], [6]. Our actual OCR detectability is around 97% at best. Our system starts with a monochrome VGA image acquisition of the upper left section of a PCB, using a NI-1752 smart camera, with a full resolution of 640x480 pixels with a maximum data transfer of 60 fps using a GigE port. The selected resolution and data transfer speed parameters meet the factory production schedule of inspected PCBs. The camera has a grayscale output image type with a maximum character resolution to cover the entire PCB characters positions, as shown in Fig. 1.



Fig. 1. PCB with no damaged characters.

As mentioned before, due to problems with previous processes in the production line, some PCBs present some residual dirt over the characters, making some characters unreadable for the pattern matching technique, as shown in the following Fig. 2.



Fig. 2. PCBs with evident residual dirt over characters.

# *Outlier Mining of a Vision Sensing Database for SVM Regression Improvement*

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**Abstract**— A 3D spatial measurement system has been enhanced by computational intelligence. The measurement system is based in opto-electronic scanning instrumentation for industrial task, robot navigation, medical scanning, and structural health monitoring applications. This paper presents new research performed in the data processing of a vision sensing database. Multivariate outlier analysis has been implemented by Mahalanobis distance in order to improve Support Vector Machine (SVM) regression algorithm results. Measurement error regression data has been used for spatial 3D measurements error correction. Demonstrating that optical systems for measurements that present non-linear behavior could be positive impacted by computational intelligence.

**Keywords**— *opto-electronics, outliers, pattern recognition, SVM, vision sensing.*

## I. INTRODUCTION

Machine vision methods to provide 3D spatial coordinates measurement have developed in a wide range of technologies for multiples fields of applications such as industrial task, robot navigation, medical scanning, and structural health monitoring. Each technology with specified properties that could be categorized as advantage and disadvantage according its utility to the application purpose. But all of them giving origin to databases with a big size and well organized collection of data, which its structure and variables selection are designed to model aspects of reality of the machine vision system, its environment and the object under monitoring, in a way that supports measurement processes and the monitoring required information. This paper presents the analysis of a vision sensing database obtained from an opto-electronic scanning system, through data acquisition and its electronic and digital processing handling, as a continuity research previously published [1-3], regarding the accuracy increase of those optical systems which presents a non-linear behavior throughout all the scanning scene. Leading to the requirement of computational intelligence techniques support, for pattern recognition to produce an enhanced output in a modern approach to the prediction of the non-linear measurement error of the systems to compensate it, as is demonstrated in

related work [4], contributing this experimentation in the regressing modelling of vision sensing databases.

## II. OUTLIER MINING

Outliers has been defined as observation or subset of observations which appears to be inconsistent with the remainder of a dataset. Outliers have different sources: they may be the result of an error, they may arise from the inherent variability of the dataset (i.e. extreme values from the tails of the distribution) or they may be generated from another model. Include or exclude outliers from a dataset could change the results of data analysis, the decision to assess this membership status depends on the reason why the instance is an outlier and the purpose of the analysis. The outliers describe the abnormal data behavior, i.e. data which are deviating from the natural data variability. The cut-off value or threshold which divides anomalous and non-anomalous data numerically is often the basis for important decisions. Concluding that outliers are instances of a particular situation that have data values that are very different from the data values for the majority of instances in the data set.

### A. Univariate Outliers

Univariate outliers are instances that have an unusual value for a single variable. Usually the dependent variable. Many methods have been proposed for univariate outlier detection. They are based on robust estimation of location and scatter, or on quantiles of the data [5]. One method to identify univariate outliers is to convert all of the scores for a variable to standard scores. The standard score indicates how many standard deviations a data is from the mean. Thus, a positive standard score indicates a data above the mean, while a negative standard score indicates a data below the mean. It is a dimensionless quantity obtained by subtracting the population mean from an individual raw score and then dividing the difference by the population standard deviation, as shown in (1).

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Where:  $z$  is the standard score,  $x$  is the raw score,  $\mu$  is the mean of the population,  $\sigma$  is the standard deviation of the



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## CONISOFT 2015 notification for paper 35

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Wed, Apr 15, 2015 at 7:02 AM

To: "Félix F. González-Navarro" <fernando.gonzalez@uabc.edu.mx>

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Nos da mucho gusto informarle que su artículo:

Modelación Matemática de la Respuesta de un Biosensor de Glucosa Oxidasa de Segunda Generación mediante Técnicas de Reconocimiento Estadístico de Patrones

ha sido ACEPTADO para el Consorcio de Tesis a realizarse en el marco del CONISOFT 2015.

Su artículo ha sido evaluado se considera que su trabajo ofrece una contribución valiosa al evento. El Consorcio de Tesis proporcionará la oportunidad para los estudiantes de maestría y doctorado de desarrollar sus ideas de investigación, aprender sobre los retos de conducir una investigación y de construir una carrera académica exitosa en el campo de la Ingeniería de Software y disciplinas relacionadas, así como expandir sus conexiones con la red nacional e internacional en el campo de investigación.

Gracias por contribuir a que el CONISOFT 2015 sea un evento exitoso que propicie el intercambio de conocimiento y las colaboraciones entre sus participantes. Nos vemos en San Luis Potosí.

Saludos cordiales.

There are no reviews for this paper



# Modelación Matemática de la Respuesta de un Biosensor de Glucosa Oxidasa de Segunda Generación mediante Técnicas de Reconocimiento Estadístico de Patrones

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## Resumen

*Un biosensor es un instrumento o dispositivo analítico compacto que contiene un elemento de detección biológico o biorreceptor y está acoplado a un transductor físico-químico que convierte la señal biológica en una señal eléctrica. Esta señal es proporcional al compuesto o grupo de compuestos que se quiere determinar y puede recoger información cualitativa o cuantitativa. El impacto potencial de los biosensores en diversos campos de la ciencia los ha hecho elementos muy prometedores dadas sus características de simplicidad, tiempo de respuesta rápido y de bajo costo. Las aplicaciones abarcan diversas áreas de investigación y desarrollo como el sector médico y el cuidado del medio ambiente, entre otros. Sin embargo, esta tecnología está todavía en desarrollo. De esta manera, en el trabajo de tesis se ofrecen contribuciones al avance del conocimiento en este campo mediante la aplicación de métodos computacionales para modelar su funcionamiento.*

## 1. Área de la Ingeniería de software

Este trabajo se desarrolla en el programa de posgrado de MyDCI-UABC área de conocimiento Computación. Las áreas de conocimiento del SWEBOK que son aplicadas en esta investigación son:

- AC\_03. Construcción de software.
- AC\_13. Fundamentos de computación.
- AC\_14. Fundamentos matemáticos.
- AC\_15. Fundamentos de ingeniería.

## 2. Planteamiento del problema

La Diabetes Mellitus (DM) es una enfermedad grave, crónico degenerativa donde los pacientes

afectados presentan altos niveles de glucosa en la sangre. Esto debido a dos principales causas:

- 1) Una producción insuficiente de insulina;
- 2) Mal funcionamiento en un tipo de células llamadas islotes.

Estas últimas no responden adecuadamente a la insulina en el proceso de regulación de la glucosa en la sangre. Cuando un paciente es diagnosticado con la enfermedad Diabetes Mellitus todo el proceso presenta una dinámica errática [1].

Debido al incremento de las cifras de pacientes que presentan DM en los últimos años, el interés por el constante monitoreo de los niveles de glucosa en la sangre (glucemia) ha ido en aumento [2]. Una de las opciones para tal efecto es el uso de biosensores, debido a su bajo costo de fabricación, alta sensibilidad y confiabilidad. Sin embargo, este campo aun se encuentra bajo desarrollo, sobre todo los biosensores de glucosa oxidasa. Aunque las pruebas con biosensores son sencillas y no requieren de gran cantidad de instrumentación, aun persisten los mecanismos de prueba y error, el cual puede ser simplificado mediante el uso de modelos matemáticos que contribuyan a su diseño y desarrollo.

El uso actual de los biosensores se ha extendido a varios campos de la ciencia y la ingeniería. La detección de contaminantes de los recursos hídricos, detección de agentes patógenos en medicamentos y en la industria alimentaria, por mencionar algunos ejemplos. En un escenario normal, un paciente requiere un examen físico y pruebas de laboratorio que tardan algunos días en obtener los resultados necesarios para un correcto tratamiento. Tal demora a veces puede derivar en complicaciones debido a la falta del tratamiento médico adecuado [1]. Es por ello que los biosensores se convierten en un dispositivo tecnológico muy prometedor en aplicaciones de salud pública. Por ello, esta tesis busca contribuir en el desarrollo de este nuevo campo aportando conocimientos en el modelado



The Mexican Society for Artificial Intelligence (SMIA) and  
the Instituto Tecnológico de Tuxtla Gutiérrez (ITTG)  
award this certificate to



**FÉLIX F. GONZÁLEZ NAVARRO**

for serving as a session chair in the parallel session

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# La Exteriorización de Activos de Conocimiento en Mejora de Procesos Software

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**Abstract**—El conocimiento es el activo más valioso en una organización de desarrollo de software. En este documento se expone la importancia de identificar, caracterizar y exteriorizar los tipos y activos de conocimiento en la institucionalización de procesos software, de tal manera que las Micro, Pequeñas y Medianas empresas (MIPyMEs) se apoyen de ellos para ser generadoras de nuevo conocimiento o proyectos de conocimiento, llevándolas así a elevar su madurez organizacional. La contribución de este trabajo es presentar un marco que ayuda a estandarizar el proceso de exteriorización de tipos, fuentes y activos de conocimiento involucrados en un proyecto de mejora de procesos software. Los resultados de la evaluación inicial sobre el marco muestran un efecto positivo en los roles sobre la planeación y realización de las actividades.

**Keywords**—mejora de procesos software; activos de conocimiento; exteriorización de conocimiento

## I. INTRODUCCIÓN

En una organización, una forma de detectar o representar su conocimiento es por medio de los activos de conocimiento, los cuales se refieren a las lecciones aprendidas, buenas o mejores prácticas, reglas heurísticas y la generación continua de ideas o experiencias de su recurso humano [1]. Las personas interiorizan y utilizan los activos de conocimiento para desempeñar sus actividades diarias, permitiéndoles reducir recursos y defectos, agregar innovación a sus productos y procesos [2, 3]. Es por esto que en las organizaciones desarrolladoras de software, se recomienda la reutilización de activos de conocimiento como una técnica útil y eficiente para exteriorizar y transmitir el conocimiento tácito que posee cada uno de los miembros del equipo de desarrollo de software, con el objetivo de mejorar la calidad de los productos, las fases de ciclo de vida del software y/o facilitar la reutilización de productos de trabajo (PT) [2].

Autores como [4, 5] hicieron énfasis en las implicaciones que se derivan de la naturaleza tácita del conocimiento debido a que: a) por su característica es difícil de imitar o sustituir, convirtiéndolo en un recurso crítico para la organización; b) el no documentarlo o hacerlo explícito, limita la capacidad de reutilizarlo; y, c) existe un alto riesgo de desaprovechar o perder el conocimiento que posee una persona cuando deja de colaborar con los demás miembros del equipo de trabajo. Esta situación crea un especial interés en el análisis de procesos centrados la adquisición, transferencia y utilización del conocimiento existente y la creación de nuevo conocimiento a partir del conocimiento tácito [6, 7].

En un proyecto de mejora de procesos software (SPI por sus siglas en inglés de *Software Process Improvement*), se crea conocimiento en proporción al nivel de madurez organizacional, puesto que los procesos de software se basan en los activos de conocimiento de las actividades realizadas [8]. Algunos estudios recientes citan proyectos de SPI en los que se resalta que el éxito se debió a la captura del conocimiento tácito y explícito existente en la organización [9, 10, 11]. De esta forma, se considera que la Gestión del conocimiento (GC) relacionada a proyectos de SPI, promueve la mejora continua de los procesos y métodos y enfatiza la generación y utilización del conocimiento relevante [8].

Este trabajo se enfoca en el proceso de exteriorización, considerando que una MIPyME dedicada al desarrollo y mantenimiento de software, que desee implementar un proyecto SPI basado en conocimiento, requerirá extraer o exteriorizar la mayor parte del conocimiento tácito de sus roles, representado en los activos de conocimiento. Dicho conocimiento le es útil en un proceso de aprendizaje intensivo de conocimiento organizacional y al plantear estrategias de éxito para lograr el nivel de competitividad deseado [12].

# CAMBIO CLIMÁTICO Y CAMBIO GLOBAL

S. Fernández-Montes  
F. S. Rodrigo



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## MODELACIÓN DE TEMPERATURAS DEL AIRE A 850 MB EN EL CASO DE LA ONDA CÁLIDA DE JULIO DEL 2006 EN EL NOROESTE DE MÉXICO

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### Resumen

El estudio de las ondas de calor se enfrenta al reto de conocer los elementos clave que intervienen en su formación y desarrollo y la manera en que afectan a la salud de la población, lo que en un futuro permitiría su modelación predictiva. Se analiza el caso de una onda cálida que afectó a la ciudad de Mexicali, Baja California, México, en los días del 16 al 26 de julio de 2006 que causó graves estragos en la población. Se identifican las principales variables que causan el desarrollo de una onda cálida, y se propone la modelación de la temperatura del aire a 850 mb ( $T_{aire_{850}}$ ) mediante un modelo de regresión múltiple (MRM) y regresión con redes neuronales (RRN). Se encuentra que el MRM explica a la  $T_{aire_{850}}$  un 67%, siendo la altura geopotencial a 500mb y la presión atmosférica al nivel del mar las variables que más contribuyen a esta explicación. En cuanto a la RRN muestra un desempeño más aceptable i.e. coeficientes de regresión de 0.89; el p-value obtenido fue de 0.0019, por lo que se puede afirmar que un modelo no lineal como la Red Neuronal tiene un mejor desempeño en la predicción de valores independientes que un modelo lineal.

**Palabras clave:** Onda cálida, temperatura del aire, regresión múltiple, redes neuronales.

### Abstract

The study of heat waves is challenged to know the key elements involved in their formation and development, and how they affect the health of the population, which in the future would allow its predictive modeling. The case of a heat wave that affected the city of Mexicali, Baja California, Mexico, on days 16 to July 26, 2006 causing serious damage on the population is analyzed. Major variables that cause the development of a heat wave are identified, and the modeling of air temperature at 850 mb ( $T_{aire_{850}}$ ) is proposed by a multiple regression model (MRM) and regression neural network (RNN). It was found that the MRM explained to  $T_{aire_{850}}$  a 67%, with a 500mb geopotential height and atmospheric pressure at sea level the atmospheric variables that more contribute to this explanation. Insomuch as RNN show a performance more, i.e. regression coefficients of 0.89; the p-value obtained was 0.0019, so we can say that a nonlinear model such as neural network has a better performance in predicting independent values that a linear model.

Key words: Heat wave, air temperature, multiple regression, neural network

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# *Machine Vision supported by Artificial Intelligence*

## *Applied to Rotatory Mirror Scanners*

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**Abstract**— A performance evaluation of different artificial intelligence methods for machine vision using a rotatory mirror scanner is presented. This assessment concludes importance results, in order to properly select a method for the development of an precise optical scanning system for machine vision, with application in Structural Health Monitoring and Robot Navigation task.

**Keywords**—Artificial intelligence; machine vision; rotatory mirror scanner, SVM.

### I. INTRODUCTION

Machine Vision is the technology and methods used to provide imaging-based automatic inspection and analysis for such applications as medical diagnostics, industrial automatic inspection, robot navigation and structural health monitoring, among others. It is related to an electromagnetic spectrum that could be sensed by a detector device.

The computer vision has guide the machine vision to the tendency of duplicate the abilities of human vision by electronically perceiving and understanding an image with a high dimensional data, increasing the data storage requirement, and the time processing due to the complexity of algorithms to extract important patterns and trends to understand what data says (learning data).

For some applications it is not necessary to acquire the whole vision of a scene as makes the human vision to extract the features of interest, it is possible to only extract the significant characteristics from the scene by the use of a rotatory mirror scanner.

A rotatory mirror scanner is used to deflect a light beam by means of a rotatory mirror driven mechanically. The rotatory mirror is mounted on a rotatory shaft on which a plate and a rotor of a motor are mounted. The rotor produces a torque by virtue of the revolving magnetic field of a stator of the motor and the rotatory mirror rotates at a predetermined number of revolutions. By this arrangement, the rotatory mirror scanner can rotate very quietly at very high speeds [1].

A rotatory mirror scanner offers the advantage of providing a larger deflection angle. However since the deflection rate (or frequency) may vary depending on the number of revolutions

of the mirror. It is necessary that the mirror should be rotated at the highest number of revolutions possible or tens of thousands of revolutions per minute in order to obtain a high resolution.

Some disadvantages are associated with a rotary mirror scanner although it rotates smoothly. When stationary, the rotatory mirror scanner experience difficulty in initiating the rotation of the rotary shaft, the motor presents eccentricity at low speed scanning [2], that could be minimized and control by electronics that provides, for a desired angle or phase, the suitable electrical current to the motor. Besides of some disadvantage due to optical effects such as reflection, diffraction, absorption and refraction, that could be minimized and control by taking the measurements at the energy centre of the optoelectronic signal by power spectrum centroid [3], but still producing a non-linear behavior that is observed on the measurement error analysis and could be compensating efficiently by the use of artificial intelligence.

The scope of the present paper is the assessment of seven artificial intelligence algorithms to select the one that best suit the data model with the goal to predict the value of the outcome error measure based on a number of input measures to compensate the error measurement caused by the few disadvantage on rotatory mirror scanners and explore the big benefits of advantages as simplicity, low cost, suitable to any kind of coherent or incoherent light detection and focus in the search of only the features of interest.

### II. SUPERVISED LEARNING METHODS UNDER ASSESSMENT (REGRESSION)

Supervised learning is a type of machine learning algorithm that uses a known dataset (called the training dataset) to make predictions. The training dataset includes input data and response values. From it, the supervised learning algorithm seeks to build a model that can make predictions of the response values for a new dataset. A test dataset is often used to validate the model. Using larger training datasets often yield models with higher predictive power that can generalize well for new datasets. Supervised learning includes two categories of algorithms: a) Classification; for categorical response values, where the data can be separated into specific “classes” and b) Regression; for continuous-response values [4].

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# Non-deterministic local search methods for feature selection. An experimental study

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**Abstract**—The dimensionality reduction by feature selection is one of the fundamental steps in the pre-processing data stage in the intelligent data analysis. Feature selection (FS) literature embodies a wide spectrum of algorithms, methods and strategies, but mostly all fall into two classes, the well known wrappers and filters. The decision of which feature or variable is selected or discarded from the best current subset is still subject of research nowadays. In this paper, an experimental study about non-deterministic local search methods as main engine to this decision making is presented. The Simulated Annealing Algorithm, the Genetic Algorithm, the Tabu Search and the Threshold Accepting Algorithm are analyzed. They are used to select subset of features on several real and artificial data sets with different configurations –i.e. continuous and discrete data, high-low number of cases/features– in a wrapper fashion. The Nearest Neighbor Classifier, the Linear and Quadratic Discriminant Classifier, the Naive Bayes classifier and the Support Vector Machine are evaluated as the performance function in the wrapper scheme.

**Keywords**—Intelligent Data Analysis; Feature Selection; Simulated Annealing; Genetic Algorithm; Tabu Search; Threshold Accepting;

## I. INTRODUCTION

The *Feature Selection* (FS) problem is ubiquitous in an inductive machine learning setting and its importance is beyond doubt. The generic purpose pursued is the improvement of the inductive learner, either in terms of learning speed, generalization capacity (defying the curse of dimensionality to enhance the prediction performance) or simplicity of the representation. When many *unuseful*, i.e. irrelevant or redundant, features are present in the training data, classifiers are prone to finding false regularities in the data and therefore are more likely to overfit. The main purpose of FS is therefore to find a reduced set of features from a data set described by a feature set. This is often carried out with a *subset search process* in the power set of possible solutions, guided by the optimization of a user-defined *objective function*.

In this paper, some non-deterministic local search strategies are explored. The Simulated Annealing Algorithm (SA), the Genetic Algorithm (GA), the Tabu Search (TS) and the Threshold Accepting Algorithm (TA) are recurrently used in a wide spectrum of scenarios and problems. Their power

to find solutions, despite its known speed problems, often surpass deterministic solutions. A number of experiments on real and artificial datasets take place in order to benchmark these four well known optimization algorithms as engines to generate subsets of features.

## II. THE FEATURE SELECTION SCENARIO AND NON-DETERMINISTIC SEARCH STRATEGIES

The feature selection field is divided into two main categories: 1) The wrapper approach, which depends on an inductor (a classifier) to determine the discriminating ability (with the disadvantage of high computational cost) and 2) The filter approach, where the process is based on the data independently of an inductor. Under this view, the FS process must possess two main elements:

- 1) The evaluation function for judging whether a subset is better (more relevant) than another.
- 2) The search strategy, which decides how to keep exploring new solutions.

The FS can be cast as a search problem, where each state in the search space corresponds to a subset of features. If the feature selection problem is seen as an open search in a hypothesis space (the set of possible solutions), then the number of potential subsets to evaluate is exponential w.r.t. the number of features. A *complete* search (as in the *branch and bound* method), is a possible procedure to guarantee the finding of an optimal subset; this method also requires the monotonicity of the inducer evaluation. This implies that when a feature is added to the current subset, the value of the criterion or evaluation function does not decrease. In most practical applications this approach is computationally prohibitive and the mainstream of research on FS has thus been directed to *deterministic sequential* suboptimal search methods. In particular, a *sequential feature selection algorithm* is a polynomial-time computational solution that is motivated by a certain definition of *relevance*.

On the other hand, non-deterministic local search methods borrow known ideas from other disciplines to construct complex search strategies which, despite their known speed performance problem, yield competitive results. In the next section, the four search strategies under analysis are explained.

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# Digital Signal Processing on Optoelectronic for SHM

W. Flores-Fuentes, M. Rivas-Lopez, O. Sergiyenko, F. Gonzalez-Navarro J. Rivera-Castillo, and D. Hernandez-Balbuena

**Abstract—** *This paper proposes a particular advantage of Support Vector Machine (SVM) to predict measurements errors for Accuracy Enhancement in Optical Scanning Systems, for position detection in real life application for Structural Health Monitoring (SHM) by a novel method, based on the Power Spectrum Centroid Calculation in determining the energy center of an optoelectronic signal. The Energy Signal Centre is found in the Power Spectrum Centroid and the SVM Regression Method is used as a digital rectified to increase measurement accuracy for Optical Scanning System.*

**Index Terms—** Support Vector Machines, Power Spectrum Centroid, Energy Signal Centre, SHM.

## I. INTRODUCTION

SHM is a technology focused on data acquisition and analysis regarding man-made structures. The data is obtained from non-destructive sensors and its analysis allows the detection of anomalies to evaluate its health [1].

In this paper, an Optical Scanner System (OSS) is subject to be enhanced by increasing its accuracy by modeling possible errors in position measurements by means of digital processing techniques.

The Power Spectrum Centroid and Support Vector Machine techniques are introduced in this research to process the OSS optoelectronic output signal in order to calculate the energy centre of the signal by Power Spectrum Centroid and to use error approximation functions to perform the digital rectification of systematic and random measurement error by using a well know machine learning regression model, the Support Vector Machine (SVM).

In an OSS system, right after data acquisition stage takes place; a data management process is carried. It consists of five steps as shown in Figure 1.

Step 1. Data processing through Fourier Transformation to translate data from time domain to frequency domain.  
Step 2. Power Spectrum computation in order to represent the data in power per frequency unit.  
Step 3. Centroid calculation, which corresponds to the energy centre of the signal (in frequency domain).  
Step 4. Convert this frequency domain coordinate to time domain coordinate to obtain the position where the light emitting source under scanning is positioned.  
Step 5. A digital rectification in order to obtain accuracy enhancement.

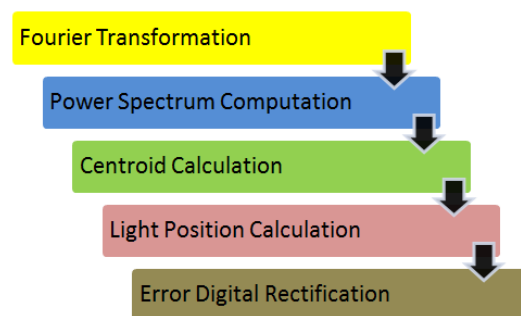


Fig. 1. Digital signal processing flow.

## II. OVERALL OSS FUNCTIONING

In this chapter, an OSS [2-4] will be described in detail. Theoretical Principles, measurements concepts and the Power Spectrum Centroid Computing Method will be described, as long as the accuracy enhanced proposal.

### A. Scanning

The OSS of this study (a 45° Sloping Cylindrical Mirror Scanner Aperture) shown in Fig 2 consists of the following elements:

1. An incoherent light source emitter (non-rotating) mounted on the structure under monitoring. That could be placed at different angles to detect its position as shown in Figure 2 at 90°, 120°, and 150°.
2. A passive rotating optical aperture sensor designed with a 45° sloping mirror, and embedded into a cylindrical micro rod. The beam of light is deviated by the 45° sloping mirror to a double convex lens and filtered, in order to remove any interference and enhance the focus.

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# Microarray Gene Subset Selection in Amyotrophic Lateral Sclerosis Classification

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**Abstract**—Amyotrophic Lateral Sclerosis (ALS) is a neurodegenerative disease causing a progressive loss of motor neurons. The disease prevalence is 5 per 100,000 people. There is no cure and it leads generally to death from respiratory failure in approximately 3-5 years after the first symptoms. The exact causes of the disease are still unknown, however, almost 20% of the known cases have shown gene mutations. The use of gene expression analysis is a powerful tool to discover the most relevant genes in a cellular process, but the high dimensionality of the data makes the feature selection a challenging task. Using a filter method combined with machine learning algorithms, an ALS data set is explored. Bootstrap resampling is used as a way to achieve stability in the whole process.

**Keywords**—Amyotrophic Lateral Sclerosis, Microarray Gene Expression Data, Gene Subset Selection, Machine Learning, Classification, Bootstrap Resampling.

## I. INTRODUCTION

Amyotrophic Lateral Sclerosis (ALS) is a neurodegenerative disease causing a degeneration and loss of upper motor neurons and the lower motor neurons. The illness is relentlessly progressive, leading to death from respiratory paralysis; having a median survival of 3.5 years. In most societies there is a prevalence of 1 - 9 per 100,000 people. In the United States and Europe, males are somewhat more frequently affected than females [1].

For years, experts have tried to find factors common to people who develop ALS, such as environmental toxins, occupational hazards, places of work or residence, and so forth. So far, the evidence for such risk factors and triggers has been frustratingly unclear [2]. Nevertheless, epidemiologic studies have incriminated risk factors for this disease including exposure to pesticides and insecticides, smoking and, in one report, service in the military. While ALS is overwhelmingly a sporadic disorder, some 10 - 20% of cases are inherited as an autosomal dominant trait [1]. Recent advances in ALS research showed two interesting genes, FUS/TLS and TDP-43, that function together and their combined activities may be disrupted in ALS—see [3].

Since ALS research in the past years have pointed out that the main source of the problem is genomic, in this paper ALS experimental findings on a microarray gene expression data set by means of Machine Learning techniques (ML) is presented. A simple Feature Selection Algorithm (FSA) being feed by a ranked list of genes is used to select relevant features or genes. Two classifiers are mainly tested, the

Naive Bayes and the Nearest Neighbor classifier. In order to guide—and hopefully— stabilize performance readings, bootstrap resampling is solicited. Experimental results show acceptable values jointly with a considerable reduction of final subset.

It is convenient to mention that during the phase of looking for evidence of similar works in this particular disease, no ML studies have been found; contrary to the existence of several ones in a wide spectrum of cancer diseases—see [4]—. Therefore, it is believed that the results arisen in this paper become a good opportunity to contribute to the scientific knowledge by discovering important genes to unlock the possibility of better ways to detect and treat the ALS. This last issue encourage us to conduct an in-depth exploration into this promising field.

The paper is organized as follows: the reader is briefly introduced to the feature selection problem and how this computer science field has been used in microarray domain. Following, the experimental study is described in detail. The bootstrap method is presented and properly discussed as long as the experimental settings. Finally the experimental results are presented and complemented with medical evidence about the most important findings.

## II. AN EXPERIMENTAL STUDY

The *feature selection* (FS) problem is ubiquitous in an inductive machine learning setting and its importance is beyond doubt [5]. The FS problem in terms of supervised inductive learning is: given a set of candidate features select a subset defined by one of three approaches: a) the subset with a specified size that optimizes an evaluation measure, b) the subset of smaller size that satisfies a certain restriction on the evaluation measure and c) the subset with the best commitment among its size and the value of its evaluation measure. The generic purpose pursued is the improvement of the inductive learner, either in terms of learning speed, generalization capacity (defying the curse of dimensionality to enhance the prediction performance) or simplicity of the representation. When many *unuseful*, i.e. irrelevant or redundant, features are present in the training data, classifiers are prone to finding false regularities in the data and therefore are more likely to overfit.

FS becomes specially notorious in application domains characterized by hundreds or thousands of features and only



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## FEATURE SELECTION IN PROTON MAGNETIC RESONANCE SPECTROSCOPY DATA OF BRAIN TUMORS

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**Keywords:** Cancer research, Feature Selection, Classification.

**Abstract.** In cancer diagnosis, classification of the different tumor types is of great importance. An accurate prediction of different tumor types provides better treatment and may minimize the negative impact of incorrectly targeted toxic or aggressive treatments. Moreover, the correct prediction of cancer types using non-invasive information –e.g.  $^1\text{H}$ -MRS data– could avoid patients to suffer collateral problems derived from exploration techniques that require surgery. A Feature Selection Algorithm specially designed to be use in  $^1\text{H}$ -MRS Proton Magnetic Resonance Spectroscopy data of brain tumors is presented. It takes advantage of a highly distinctive aspect in this data: some metabolite levels are notoriously different between types of tumors. Experimental readings on an international dataset show highly competitive models in terms of accuracy, complexity and medical interpretability.

### 1 Introduction

Proton (or Hydrogen) Magnetic Resonance Spectroscopy ( $^1\text{H}$ -MRS) has been used extensively in biochemistry for *in vitro* chemical analysis of small samples for several years. As a technique for *in vivo* sampling of biological tissue, it provides a quantified biochemical fingerprint of metabolite concentrations [1].  $^1\text{H}$ -MRS data has the appearance of a plot of peaks along the x-axis, with the peak position depending on the resonant frequency of the associated metabolite [2]. An example of a  $^1\text{H}$ -MRS dataset is shown in Fig. 1.

Nowadays,  $^1\text{H}$ -MRS has been proven its value as a powerful tool in the clinical assessment of several pathologic conditions -e.g. epilepsy, multiple sclerosis, cancer– and specially in neurological affections [3, 4]. Framed as a minimal invasive technique, its application in brain tumor oncological diagnosis carries tremendous benefits to patients, relieving them from complicated surgical procedures and minimizing trauma to normal tissue surrounding the particular lesion or other vital elements.

The use of systematic approaches based on  $^1\text{H}$ -MRS data for the diagnosis and grading of adult brain tumors is subject of an extensive scientific research. One of these growing approaches takes as backbone well-established machine learning techniques to develop predictive models able to discern between several classes of brain tumors [5, 6, 7]. This particular task is quite challenging, mainly because of the high dimensionality and relatively low number of observations. Therefore, the use of *dimensionality reduction* (and, in particular, feature selection) methods becomes an option in order to present low complexity and interpretable models to oncologists.

In this study, an *ad hoc* feature selection (FS) algorithm is used to generate relevant subsets of spectral frequencies (considered as features). The developed algorithm takes

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## Differentiation of Glioblastomas and Metastases using $^1\text{H}$ -MRS spectral data

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**Abstract**—Hydrogen-1 magnetic resonance spectroscopy ( $^1\text{H}$ -MRS) allows noninvasive *in vivo* quantification of metabolite concentrations in brain tissue. In this work two of the most aggressive brain tumors are studied with the purpose of differentiating them. The challenging aspect in this task resides in that their radiological appearance is often similar, despite the fact that treatment of patients suffering these conditions is quite different. Efforts to differentiate between these two profiles are getting increasing attention, mainly because the consequences of performing an incorrect diagnosis. Due to the high dimensionality, initiatives oriented to reduce the description complexity become important. In this work we present a feature selection algorithm that generates relevant subsets of spectral frequencies. Experimental results deliver models that are both simple in terms of numbers of frequencies and show good generalization capabilities.

**Keywords:** Feature Selection; Proton Magnetic Resonance Spectroscopy; Classification.

### 1. Introduction

Hydrogen-1 ( $^1\text{H}$ ) magnetic resonance (MR) spectroscopy allows noninvasive *in vivo* quantification of metabolite concentrations in brain tissue. Nowadays,  $^1\text{H}$ -MRS has proven its value as a powerful tool in the clinical assessment of several pathologies—as epilepsy, multiple sclerosis and several types of cancer, see e.g. [1], [2]. Its application in brain tumor oncologic diagnosis carries tremendous benefits to patients, relieving them from complicated surgical procedures and their collateral difficulties.

The use of systematic approaches based on  $^1\text{H}$ -MRS data for the diagnosis and grading of adult brain tumors is subject of an extensive scientific research. One of these growing approaches takes as a backbone well-established machine learning techniques acting as model building methodologies to discern and predict several classes of brain tumors. Previous work typically focuses on discriminating normal tissue from high grade malignant tumors, low grade malignant tumors, meningiomas, as well as other rare tumors or super-classes thereof [3], [4], [5]. In this work we go a step beyond by concentrating on the specific internal separation of the

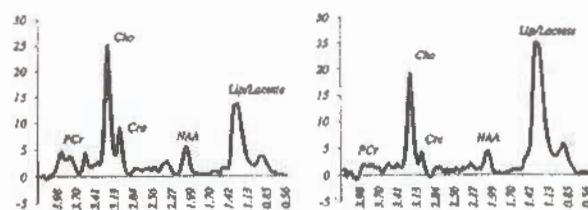


Fig. 1: Mean spectra for Glioblastomas (left) and Metastases (right).

superclass formed by the high grade malignant tumors, differentiating from Glioblastomas and Metastases. Although their radiological appearance is often similar, the clinical management of patients suffering any of these conditions is quite different [6]. This circumstance may lead to a surgical biopsy for a definitive diagnosis, an undesirable situation in general. Therefore, the use of non-invasive techniques and the discriminative power of the derived information is ever encouraged in this setting.

### 2. High-grade gliomas and Metastases

Glioblastomas (GLI) are the most common and most aggressive type of primary brain tumor in humans and account for the 52% of all parenchymal brain tumor cases and 20% of all intracranial tumors. It affects mainly male patients aged 45 to 55, with a survival expectation of around 40 weeks after diagnosis and medical treatment [7], [8].

Brain Metastases (MET), which amounts about 35% of all cancer patients, are a major cause of death from cancer. Properly speaking, METs are not originated in the Central Nervous System (CNS). Their source is mostly based in lung cancer, breast cancer, melanoma, renal cancer and colon cancer. Mutated cells are spread out through direct invasion to normal brain tissue or by alternate routes like the lymphatic and blood vessels circulating through the bloodstream. Patients with brain metastases at best reach a mean survival expectation, after whole brain radiotherapy of a little more than seven months [9], [7].



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# Using Machine Learning Techniques to Explore $^1\text{H}$ -MRS data of Brain Tumors

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**Abstract**—Machine learning is a powerful paradigm to analyze Proton Magnetic Resonance Spectroscopy ( $^1\text{H}$ -MRS) spectral data for the classification of brain tumor pathologies. An important characteristic of this task is the high dimensionality of the involved data sets. In this work we apply filter feature selection methods on three types of  $^1\text{H}$ -MRS spectral data: long echo time, short echo time and an *ad hoc* combination of both. The experimental findings show that feature selection permits to drastically reduce the dimension, offering at the same time very attractive solutions both in terms of prediction accuracy and the ability to interpret the involved spectral frequencies. A linear dimensionality reduction technique that preserves the class discrimination capabilities is additionally used for visualization of the selected frequencies.

**Keywords**—Machine Learning; Feature Selection; Proton Magnetic Resonance Spectroscopy; Classification; Visualization;

## I. INTRODUCTION

Proton Magnetic Resonance Spectroscopy or  $^1\text{H}$ -MRS is a non-invasive technique that provides information about the biochemical profile (metabolites and lipids) of brain tissue;  $^1\text{H}$ -MRS data has the appearance of a plot of peaks along the x-axis, with the peak position depending on the resonant frequency of the associated metabolite. As a result this technique provides a valuable tool in discovering or telling from possible disease processes [1] and serves as a support to other diagnostic methods like Magnetic Resonance Imaging (MRI) in clinical practice [2]. An example of a  $^1\text{H}$ -MRS data set is depicted in Fig. 1. There are several examples in the literature that use machine learning (ML) techniques in telling different classes of brain tumors. First attempts using  $^1\text{H}$ -MRS data in assessing human brain tumors *in vivo* are back to [3]. It was found that spectra differ significantly from normal brain spectra and between tumors by detecting the presence/absence of different metabolites. Even though no ML analysis of spectra was done in establishing these differences, it was concluded that  $^1\text{H}$ -MRS spectroscopy may help to differentiate tumors for diagnostic and therapeutic purposes, limiting the need for invasive and risky diagnostic procedures such as biopsies. ML techniques have been used in order to *automate* the classification task. Linear discriminant analysis (LDA) (e.g. [5], [6]) and artificial neural networks (e.g. [7], [8]) are commonly used methods. Later studies perform dimensionality reduction, either considering

the *peak* signals, ratios between peak signals or *feature extraction* [14], [13]. First studies in performing explicit feature selection used a modification of the well-known Forward Selection method to select spectral points [9]. More recently, *filter* methods such as the Fisher criterion and the Relief algorithm, among others, have been considered [2].

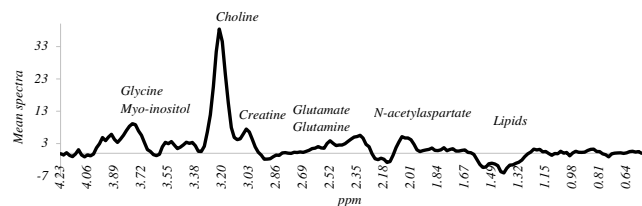


Figure 1. Example mean spectra for a long echo time  $^1\text{H}$ -MRS data set. Some metabolites –resulting products of metabolic processes– are indicated in the spectra. For example: Creatine (tCre), a marker of oxidative metabolism of cells; Choline, which is a combination of multiple metabolites and is elevated in all brain tumors; Lipids (Lip), seen in condition of necrosis.

Due to the relatively high dimensionality of  $^1\text{H}$ -MRS data (up to 500 spectral features), these efforts use dimensionality reduction methods such as PCA to lower the complexity of the problem; a recognized drawback is the difficulty to interpret the solution in original terms; moreover, PCA is not designed to preserve class separability [9]. However, data projection techniques that preserve the class discrimination achieved by a classifier are very welcome since they can be used to visualize the solutions generated by the feature selection process. In this work we develop a feature selection method to generate relevant subsets of spectral frequencies. The evaluation of subsets is based on entropic measures that treat single spectral frequencies as the features. The goal is to find attractive solutions in terms of low numbers of frequencies, high prediction accuracy and the interpretability of the selected spectral frequencies. The combination of feature selection and classification aims at obtaining simple models (in terms of low numbers of features) capable of good generalization. We also aim to progress in the comparison of MRS data acquired at different echo times, as well as in the comparison of these with data that combine both echo times.

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# Machine learning methods for classifying normal vs. tumorous tissue with spectral data

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**Abstract.** Machine learning is a powerful paradigm within which to analyze <sup>1</sup>H-MRS spectral data for the automated classification of tumor pathologies aimed to facilitate clinical diagnosis. The high dimensionality of the involved data sets makes the discover of computational models a challenging task. In this study we apply a feature selection algorithm in order to reduce the complexity of the problem. The obtained experimental results yield a remarkable classification performance of the final induced models, both in terms of prediction accuracy and number of involved spectral frequencies. A dimensionality reduction technique that preserves the class discrimination capabilities is used for the visualization of the final selected frequencies, thus enhancing their interpretability.

**Keywords:** Brain tumor classification, Feature Selection, Visualization

## 1 Introduction

Proton Magnetic Resonance Spectroscopy (<sup>1</sup>H-MRS) is a powerful technique that helps to observe metabolic processes in living tissue [1]. Although these metabolic functions are not fully understood, it is possible to employ *machine learning* (ML) techniques on this kind of data for the diagnosis and grading of adult brain tumors [2]. Several recent examples in the literature use ML techniques for distinguishing between different brain tumor pathologies (e.g. [3], [4]). Due to the high dimensionality (nearly 390 spectral points in this study), many efforts have used dimensionality reduction methods (mainly feature extraction) to lower the complexity of the problem. Methods such as Principal Components Analysis (PCA) have recognized drawbacks that limit their applicability in this problem: the result is difficult to interpret in original terms and PCA it is not designed to preserve class separability. In the present study we are interested in performing a feature selection study in two types of <sup>1</sup>H-MRS spectral data by constructing an *ad hoc* combination of both. We use a simple filter technique for feature selection as a fast method to generate relevant subsets of spectral frequencies, complemented by the use of *bootstrapping* techniques. In this paper, we report experimental results that support the practical advantage of combining robust feature selection and classification techniques in this application field.

## Feature Selection in Proton Magnetic Resonance Spectroscopy for Brain Tumor Classification

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**Abstract.**  $^1\text{H}$ -MRS is a technique that uses response of protons under certain magnetic conditions to reveal the biochemical structure of human tissue. An important application is found in brain tumor diagnosis, due to the known complications of physical exploration and as a help to other kind of non invasive methods. It is possible to analyze spectral data with machine learning methods to classify tumor classes in an automated fashion. One important characteristic of these data is their high dimensionality. In this work we present a contribution to lighten this situation with an algorithm based on entropic measures of subsets of spectral data. Experimental results show that the approach used has a good classification performance, both in terms of prediction accuracy and number of involved spectral frequencies.

### 1 Introduction

Proton magnetic resonance spectroscopy ( $^1\text{H}$ -MRS) is a non-invasive technique that provides information about the biochemical profile (metabolites and lipids) of brain tissue. Originally used for *in vitro* chemical analysis of small samples, it has been used in the diagnosis of adult brain tumors [1]. Previous existing work shows that it is possible to classify brain tumors using the values of data points of  $^1\text{H}$ -MRS using machine learning techniques [2]. Some of these efforts perform dimensionality reduction with algorithms for feature extraction or pairwise feature selection techniques [3]. In this work, an Entropic Filtering Algorithm (EFA) for feature selection is described as a method to generate a relevant subset of spectral frequencies. This is a feature selection method based on finding feature subsets evaluated as a whole with respect to their ability to classify tumors, rather than on ranking individual contribution (that implicitly denies interaction between features). The EFA is tested on real  $^1\text{H}$ -MRS data of 6 tumor classes (grouped in three super-classes). Several machine learning algorithms are used to test the reliability of the obtained subsets in the classification of tumors, within an appropriate experimental framework.

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# Un algoritmo para el cálculo de la relevancia entrópica multivariada y su uso en la selección de variables

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## Resumen

La reducción de la dimensionalidad mediante la selección de variables es uno de los pasos fundamentales del preprocesado de datos, como fase previa al análisis de información y descubrimiento de conocimiento. De entre los diversos algoritmos de reducción de la dimensionalidad, muchos se basan en dos enfoques, el *wrapper* que utiliza un inductor (e.g. un clasificador) como medida de desempeño del proceso, y el *filter*, el cual analiza los datos sin recurrir a un inductor. La desventaja del primero es el alto coste computacional que implica evaluar un clasificador cientos o miles de veces. En este artículo se presenta un algoritmo *filter* muy eficiente para la reducción de la dimensionalidad mediante el cálculo de la relevancia entrópica multivariada. Se presentan resultados comparativos experimentales en trece problemas de *benchmarking*. El algoritmo propuesto es comparado con un algoritmo de búsqueda hacia adelante tipo *wrapper*. Los resultados indican que el algoritmo propuesto encuentra soluciones iguales o mejores con un coste inferior.

## 1. Introducción

La reducción de la dimensionalidad mediante la *selección de variables* es uno de los pasos fundamentales del preprocesado de información [10]. La selección de variables es un proceso en el cual se selecciona un subconjunto de variables optimizando lo mejor posible un criterio de evaluación dado. Esta reducción es

esencial por varias razones: reduce la complejidad de la construcción de un clasificador, elimina conocimiento irrelevante o redundante y mejora la capacidad de generalización [4].

Es común plantear un problema de tipo combinatorio, en el que el espacio de búsqueda es analizado buscando soluciones subóptimas, que posean al máximo las características de la totalidad del modelo estudiado. Siendo  $\{X_1, \dots, X_n\}$  el conjunto total de variables que representan la medición de un sistema o modelo, se selecciona un subconjunto  $\{X_{i_1}, \dots, X_{i_m}\}$  con  $m < n$  e  $i_j \in \{1, \dots, n\}$ , en el que es posible encontrar más de una solución igualmente satisfactoria.

Los métodos de selección de variables son divididos en dos grandes categorías: 1) el enfoque *wrapper*, el cual depende de un inductor (e.g. un clasificador) para determinar la habilidad discriminante (con la desventaja del alto coste computacional) y 2) el enfoque *filter*, donde el proceso está basado en los datos de manera independiente de un inductor [5]. Bajo este punto de vista, el proceso de selección de variables debe contener dos elementos principales: 1) la función de evaluación, que permitirá juzgar si un subconjunto es mejor (más *relevante*) que otro y 2) la estrategia de búsqueda, que decide cómo se siguen explorando nuevas soluciones.

En el espectro computacional de técnicas y algoritmos existen diversas funciones de evaluación y estrategias de búsqueda:

1. Funciones de evaluación: descripción de conceptos mínimos [1], información mutua, conteo de inconsistencias [12], sepa-

# A Thermodynamic Algorithm for Feature Selection

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## Abstract

The main purpose of Feature Selection (FS) is to find a reduced subset of attributes from a data set described by a feature set. This implies a search process in the space of possible solutions, trying to optimize an objective function. This work introduces TAFS, a Thermodynamic Annealing Feature Selection algorithm. Given a suitable objective function, TAFS uses a special-purpose implementation of simulated annealing to find a good subset of attributes that maximizes this objective function. A distinctive characteristic of TAFS over other search algorithms for feature subset selection is its probabilistic capability to accept momentarily worse solutions. TAFS has been evaluated against one of the most robust and reliable algorithms, the Sequential Forward Floating Search method (SFFS). Our experimental results show that TAFS achieves significant improvements over SFFS in the objective function for classification tasks with a reasonable reduction in subset size.

## 1 Introduction

The importance of FS is beyond doubt in inductive machine learning and data mining [7]. The FS problem in terms of supervised inductive learning can be expressed as: given a set of candidate features, select a subset defined by one of two approaches: a) the subset with a specified size that optimizes an objective function, b) the subset of smaller size that satisfies a certain restriction on the ob-

jective function. The generic purpose pursued is the improvement of the inductive learner, either in terms of learning speed, generalization capacity or simplicity of representation. Other advantages are easier understanding of the results obtained by the inducer and reduction of the noise generated by irrelevant or redundant features, thereby eliminating useless knowledge. An important family of algorithms perform an explicit search in the space of subsets by iteratively adding and/or removing features one at a time until some stop condition is met. This process is guided by the objective function (usually the inducer itself).

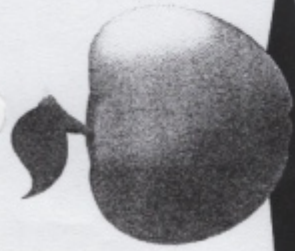
This work addresses the problem of selecting a subset of features from a given set in an inductive learning setting by introducing TAFS (Thermodynamic Annealing Feature Selection) a novel algorithm for FS. Given a suitable objective function, the algorithm makes use of a special-purpose *simulated annealing* technique to find a good subset of attributes that maximizes the objective function. The algorithm is parameterized by the initial and final temperature settings and the value of the  $\epsilon$  *parameter*. These values are necessary to achieve the desired eventual exploration of worse solutions. A further characteristic of TAFS over other search algorithms for feature subset selection is its probabilistic capability to accept momentarily worse solutions, which in the end may result in better hypotheses. The TAFS algorithm is evaluated against the *Sequential Forward Floating Search* (SFFS) proposed by [10], one of the most robust and reliable algorithms. In our ex-



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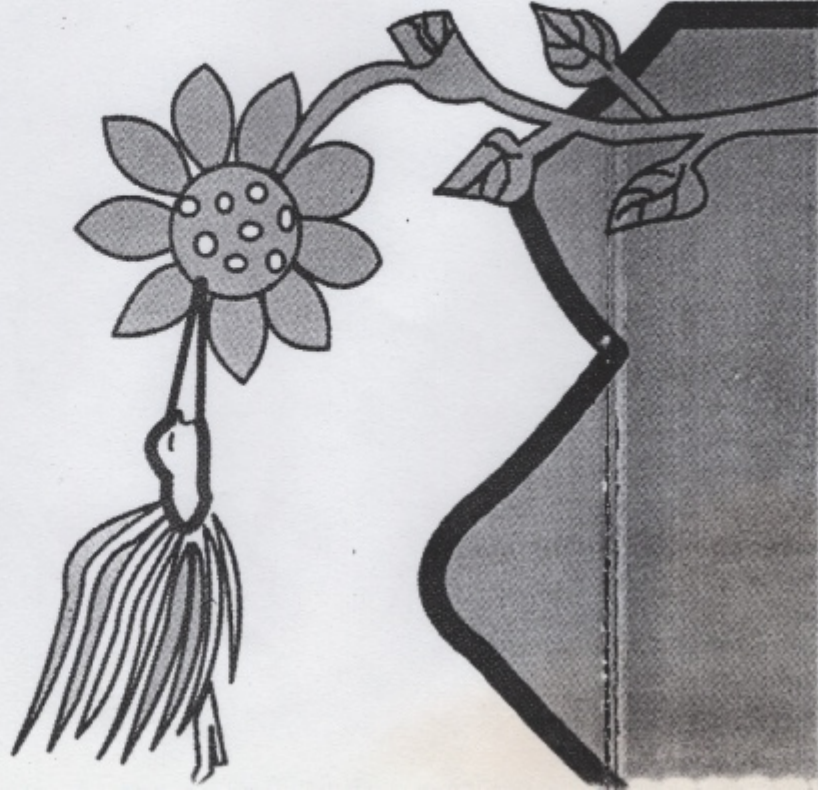
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